

Collagen Molecules

Collagen, the most abundant protein in mammals, constitutes about one-quarter of the total body protein. It occurs in skin, bones, tendons, cartilage, blood vessels, teeth, basement membranes, the corneas and vitreous bodies, and supporting tissues of the internal organs. Collagen forms interlinked, insoluble threads (fibrils) of unusual strength. A fiber of 1 mm diameter can hold a weight of almost 10 kg. More than ten distinct human diseases are caused by mutations in one of the genes encoding collagen. Collagen genes form a multigene family with more than 28 members, their genes being located on 12 different chromosomes.

A. Collagen structure

The amino acid sequence of collagen is simple and periodic (1). Every third amino acid is glycine (Gly). Other amino acids alternate between the glycines. The general structural motif is (Gly-X-Y)_n. X is either proline or hydroxyproline; Y is either lysine or hydroxylysine (2). Three chains of collagen form a triple helix (3). In collagen type I, the helix is composed of two identical α1 chains and an α2 chain. It is first formed as a precursor molecule, procollagen (4). Procollagen peptidases remove peptides at the N-terminal and C-terminal ends to form tropocollagen (5). Tropocollagen molecules are connected by the numerous hydroxylated proline and lysine residues to form collagen fibrils (6). Each fibril consists of staggered, parallel rows of end-to-end-tropocollagen molecules, separated by gaps (7). Collagen fibrils are visible as transverse stripes under the electron

microscope (8). (Photograph from Stryer, 1995.)

B. Prototype of a gene (COL2A1) for procollagen type II (α1[II])

A procollagen molecule is encoded by a gene consisting of 52 exons. The translated part of exon 1 (85 base pairs) codes for a signal peptide necessary for secretion. Exon sizes differ, with one exon coding for 5, 6, 11, 12, or 18 periodic Gly-X-Y units. The genes for procollagen types I, II, and III differ in that some exons are fused, but otherwise they are similar, especially for the three main fibrillar collagen types (I, II, III).

C. Gene structure and procollagen type α1(I)

The 52 exons of the COL1A1 gene correspond to the different domains (A to G) of procollagen α1(I). The COL1A2 gene for procollagen α2(I) is about twice as large (~40 kb) as the COL1A1 gene because the introns between the exons are on average twice as long as in COL1A1.

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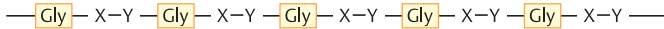
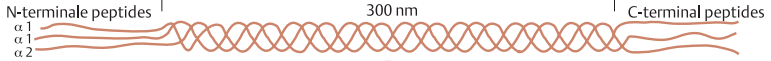
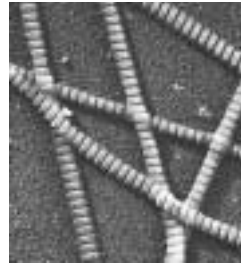
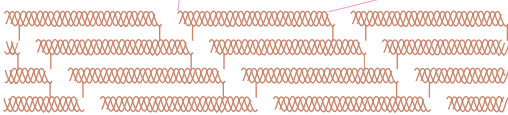
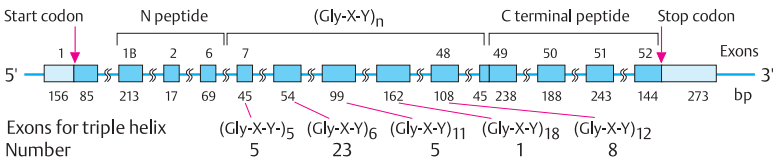
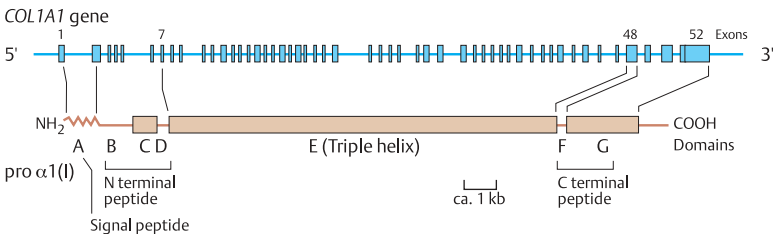
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Examples of different types of collagen (four of twenty-three)

Collagen Type	Gene	Localization	Chain	Molecule	Tissue distribution
I	COL1A1 COL1A2	17q21–22 7q21–22	α1(I)	[α1(I)] ₂ α2(I)	Skin, tendons, bone, arteries
II	COL2A1	12q13–14	α1(II)	[α1(II)] ₃	Cartilage, vitreous body
III	COL3A1	2q31–32	α1(III)	[α1(III)] ₃	Skin, arteries, uterus
IV	COL4A1 COL4A2 (6 α chains)	13q33–34	α1(IV) α2(IV)	[α1(IV)] ₂ α2(IV) [α1(IV)] ₃ [α2(IV)] ₃	Basement membranes

1. Amino acid sequence:

Glycine — Proline — Hydroxyproline — Glycine — Proline — Hydroxyproline — Glycine — Proline — Hydroxyproline —

2. General:**3. Collagen triple helix:****4. Procollagen:****5. Tropo-Collagen****6. Collagen fibril:****7. Structural pattern of a fibril****A. Collagen structure****8. Fibrils in EM****B. Prototype of a gene (COL2A1) for procollagen type II (α1[II])****C. Gene structure and procollagen type α1(I)**

Osteogenesis Imperfecta

Osteogenesis imperfecta (OI) ("brittle bone disease") is a heterogeneous group of clinically and genetically different types of diseases with a total frequency of at least 1 in 10 000 individuals. Spontaneously occurring bone fractures, bone deformity, small stature, defective dentition (dentinogenesis imperfecta), hearing impairment due to faulty formation of the auditory ossicles, and blue sclerae (the fully developed conjunctiva of the eye is thinner than normal, so that refracted light is shifted toward blue) occur in the various forms to different extents and with different grades of severity, depending on the type of mutation. OI has been demonstrated in a seventh-century Egyptian mummy (cited from Byers, 1993).

A. Molecular mechanisms in osteogenesis imperfecta

Some types of mutation may lead to reduced production of pro α 1(I) (1 and 2), e.g., deletion of a *COL1A1* allele, a transcription or splicing defect, or faulty formation of collagen fibrils. The relative excess of pro α 2(I) molecules becomes degraded. Thus, less procollagen than normal is formed, but it is not defective. Numerous other types of mutations can lead to defective procollagen (3). Mutations in the pro α 1(I) gene are more severe than mutations in the pro α 2(I) gene because a greater amount of defective collagen is formed. (Figure after Wenstrup et al., 1990).

B. Mutations and phenotype

The location of a mutation in the gene influences the phenotype. Generally, mutations in the 3' region are more serious than mutations in the 5' region (position effect). Mutations of the pro α 1(I) chain are more severe than those in the pro α 2(I) chain (chain effect). The substitution of a larger amino acid for glycine, which is indispensable for the formation of the triple helix, leads to severe disorders (size effect). Different types of mutations may occur, such as deletions, mutations in the promoter or enhancer, and splicing mutations. The codons (AAG, AAA) for the amino acid lysine, which occurs frequently in collagen, are readily transformed into a stop codon by substitution of the first adenine by a thymine (TAG or TAA), so that a short, unstable procollagen is formed. Splicing

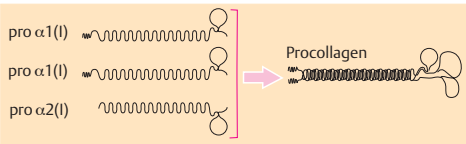
mutations may lead to the loss of exons (exon skipping). (Figure from Byers, 1990).

C. Different forms of osteogenesis imperfecta

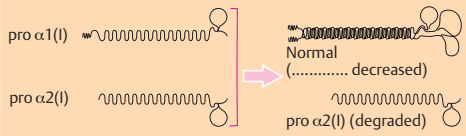
Osteogenesis imperfecta may be classified according to severity into four basic phenotypes (Sillence classification). Although the classification does not correspond to the types of mutation, it has in general proved clinically useful. OI types I and IV are less severe than type II (lethal in infancy) and type III. Three radiographs show a relatively mild (but for the patient nevertheless very disabling) deformity of the tibia and fibula in OI type IV (1); severe deformities in the tibia and fibula in OI type III (2); and the distinctly thickened and shortened long bones in the lethal OI type II (3). Mutations in OI are autosomal dominant, the severe forms being due to *de novo* mutations. Germline mosaicism has been shown to account for rare instances of affected siblings being born to unaffected parents.

References

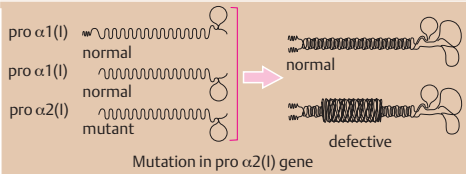
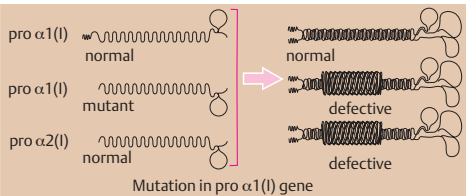
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1. Normal



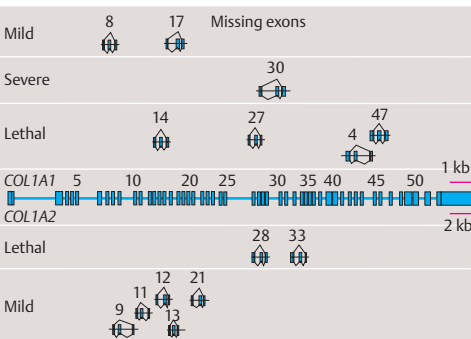
2. Decreased synthesis of procollagen $\alpha 1(I)$



3. Defective procollagen due to a mutation

A. Molecular mechanisms in osteogenesis

The position of mutations determine the phenotype



B. Mutations and phenotype



1. Bone deformation (OI type IV)



2. Severe deformation (OI type III)



3. Lethal form (OI type II)

C. Different forms of osteogenesis imperfecta

Molecular Basis of Bone Development

The skeleton develops from mesodermal cells committed to differentiate into three specialized cell types: chondrocytes (cartilage-forming cells), osteoblasts (bone-forming cells), osteoclasts (bone-degrading cells), and their precursor cells. Osteoblasts produce most of the proteins for the extracellular bone matrix and control its mineralization. The osteoblast cell lineage involves osteoblast-specific transcription factors (OSFs). One such transcription factor was identified in 1997 as a major regulator of osteoblast differentiation, the core-binding factor *Cbfa1*.

The mouse *Cbfa1* transcription factor (the human counterpart is referred to as *CBFA1*) is a member of the *runt*-domain family. The *runt*-domain is a DNA-binding domain homologous to that produced by the *Drosophila* pair-rule gene *runt*.

A. Effects of homozygous *Cbfa1* mutations on the mouse skeleton

Targeted disruption of the *Cbfa1* gene in the homozygous state ($-/-$) results in a severe phenotype. Homozygous mutant mice are small and die from respiratory failure at birth. In contrast to normal mice ($+/+$, 1), the mutant mice ($-/-$, 2) completely lack bone development, as shown by lack of alizarin red staining. Just before birth, normal $+/+$ mouse embryos (3) show well-developed bones (stained red) in the upper extremities, including the clavicles (arrow) and the tuberositas humeri (circle). Heterozygous mice ($+/-$, 4) show severe hypoplasia and reduced ossification of the long bones. Homozygous mutant mice ($-/-$, 5) totally lack any bone formation, as shown by lack of red staining.

B. Mice heterozygous for a mutation in the *Cbfa1* gene

Mice heterozygous ($+/-$) for a mutation in the *Cbfa1* gene on mouse chromosome 17 show lack of ossification of the skull (2) compared with normal mice (1). Normal calcified bone is stained red by alizarin red, here at embryonic day 17.5, three and a half days before birth. Cartilage is stained blue by alcian blue. Heterozygous mice lack a clavicle (4, arrows) in contrast to normal mice (3).

C. Cleidocranial dysplasia in humans

Cleidocranial dysplasia (CCD, McKusick 118980) is an autosomal dominant skeletal disease characterized mainly by absence of the clavicles and deficient bone formation of the skull. CCD is now considered a generalized bone dysplasia. Radiological findings are consistent with generalized underossification (Mundlos, 1999). Patients can oppose their shoulders (1) due to absence of the clavicles (2; photograph by Dr. J. Warkany, Cincinnati). The calvarium (skull case) is enlarged with a poorly ossified mid-frontal area (3). (Figure in 3 from Mundlos et al., 1997).

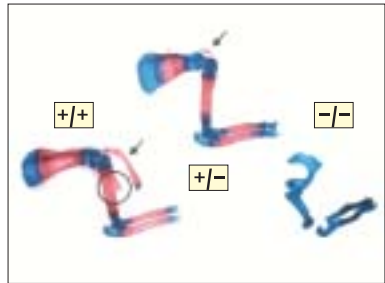
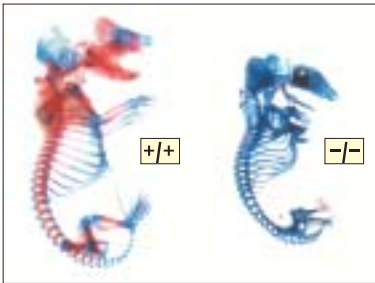
D. The human *CBFA1* gene

The *CBFA1* gene, at chromosome location 6p21, encodes a transcription factor of the core-binding factor (CBF) family. The human *CBFA1* gene has two alternative transcription initiation sites with two promoters (P1 and P2) and seven exons. Part of exon 1 and exons 2 and 3 encode the DNA-binding *runt* domain; exons 4, 5, 6, and 7 code for transcriptional activation and repression domains. The nuclear localization signal (NLS) is located at the 5' end of exon 3. Exon 6 is alternatively spliced and unique to *CBFA1*. The role of the *CBFA1* gene also includes a major regulatory function in chondrocyte differentiation during endochondral bone formation (Mundlos, 1999). As such, it functions as a "master gene" in bone development.

A wide range of mutations have been found: All mutations result in loss of function, i.e., haploinsufficiency causes the CCD phenotype. (Figures in A, B, and D kindly provided by Dr. Stefan Mundlos, Max-Planck Institute for Molecular Genetics, Berlin).

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1. Normal skeleton 2. Homozygous mutant, lack of bones 3. Normal humerus 4. Hypoplasia 5. Lack of bone development

A. Effect of homozygous *Cbfa1* mutations on the mouse skeleton



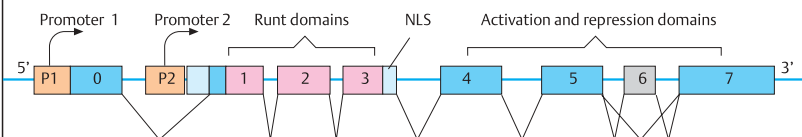
1. Normal skull 2. Lack of ossification 3. Normal thorax 4. Absent clavicles

B. Mice heterozygous for a mutation in the *Cbfa1* gene



1. Absent clavicles 2. X-ray: narrow thorax and absent clavicles 3. Lack of skull ossification

C. Cleidocranial dysplasia in human



D. The human *CBFA1* gene

Mammalian Sex Determination and Differentiation

Sex Determination

In the 1940's, the French embryologist Alfred Jost observed that when the undifferentiated gonads were removed from a male rabbit fetus before male development had begun, it developed as a female. In 1959, chromosomal analysis of two disorders in man, Turner syndrome and Klinefelter syndrome, yielded the first evidence that genetic factors on the Y chromosomes of mammals are important in determining male sex. A specific gene on the mammalian Y chromosome (*SRY*, sex-related Y) induces male sex development during embryogenesis (sex determination).

A. Determination of male phenotype by the Y chromosome

Individuals with Turner syndrome have only one X chromosome (no Y chromosome) and a female phenotype, although incompletely developed and usually accompanied by malformations. Individuals with Klinefelter syndrome have two X chromosomes, a Y chromosome, and a male phenotype, although also incompletely developed (p. 402).

B. Sex-determining region *SRY* on the Y chromosome

The relevant region in man lies in the distal short arm of the Y chromosome at Yp11.32. The short arm and the proximal half of the long arm of the Y chromosome have been divided into seven intervals (2). The most distal region of the short arm is the pseudoautosomal region 1 (PAR 1). This region is homologous to the distal segment of the short arm of the X chromosome. Homologous pairing occurs here with crossing-over during meiosis. The physical map of the pseudoautosomal region and the proximal half of interval 1 (1A1–1B) span somewhat more than 2500 kb in man (3). Intervals 2–7 contain no genes for male sex determination. The crucial portion of the Y chromosome for male sex determination in man is about 35 kb (in the mouse about 14 kb) of a region designated *Sry* (sex-related Y chromosomal region) in the in-

terval 1A1 proximal to the pseudoautosomal region (1). (After U. Wolf et al., 1992).

C. Male development of an XX mouse transgenic for the *Sry* gene

Clinical observations and experimental evidence indicate that the presence of *SRY* induces male development, irrespective of the presence of the remainder of the Y chromosome. A chromosomally female transgenic mouse (XX) shows normal male development after the 14 kb DNA fragment carrying the *Sry* region of a mouse Y chromosome is implanted into its blastocyst. (Figure from Koopman et al., 1991).

D. *Sry* expression during embryonic gonadal development of the mouse

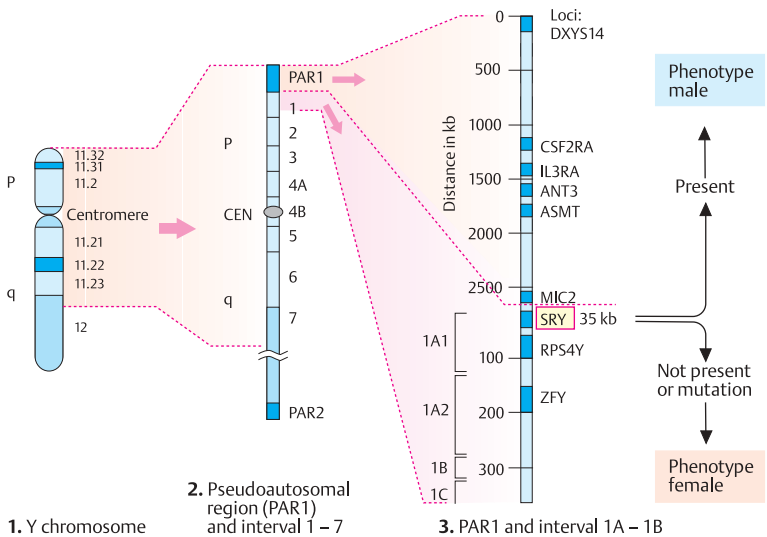
During embryonic development of an XY mouse, *Sry* is expressed only between days 10.5 and 12.5. The subsequent events leading to male development are initiated during this short time of expression. (Figure from Koopman and Gubbay, 1991).

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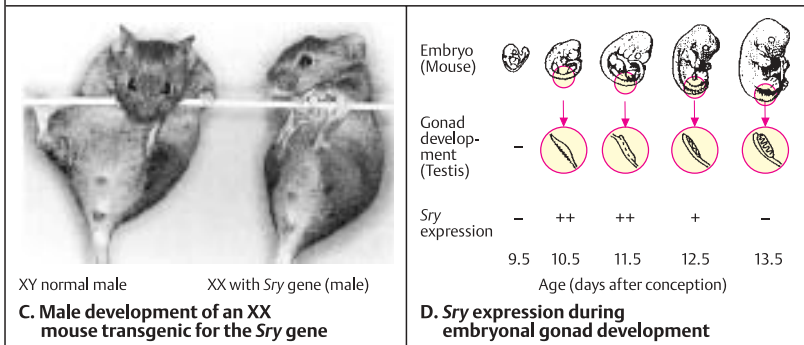
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No Y chromosome present		Y chromosome present	
45, XO One X chromosome only ↓ Turner syndrome female	46, XX ↓ Normal female	46, XY ↓ Normal male	47, XXY Additional X chromosome ↓ Klinefelter syndrome male

A. Determination of the male phenotype by the Y chromosome



B. Sex-determining region SRY on the Y chromosome



Sex Differentiation

Sex differentiation (development of a given sex) consists of many genetically regulated, hierarchical developmental steps. In mammals, the development of male structures requires induction by appropriate genes.

A. Indifferent anlagen of sex differentiation

The gonads (1), the efferent ducts (mesonephric and paramesonephric) (2), and the external genitalia (3) all develop from an indifferent stage. At about the end of the sixth week of pregnancy in humans, after the primordial germ cells of the embryo have migrated into the initially undifferentiated gonads, an inner portion (medulla) and an outer portion (cortex) of the gonads can be distinguished. When a normal Y chromosome is present, early embryonic testes develop at about the 10th week of pregnancy under the influence of a testis-determining factor (TDF). If a normal Y or TDF (SRY) is not present, ovaries develop. The wolffian ducts, the precursors of the male efferent ducts (vas deferens, seminal vesicles, and prostate), develop under the influence of testosterone, a male steroid hormone formed in the fetal testis. At the same time, the müllerian ducts—precursors of the fallopian tubes, the uterus, and the upper vagina—are suppressed by a hormone, the Müllerian Inhibition Factor (MIF; also known as anti-müllerian hormone, AMH). When testosterone is absent or ineffective, the wolffian ducts degenerate. The müllerian ducts develop under the influence of estradiol, a hormone produced by the fetal ovaries. The external genitalia (3) in humans do not develop until relatively late, starting in the 15th to 16th week. Full development of male external genitalia depends on a derivative of male-inducing testosterone, 5-dihydrotestosterone, a metabolite of testosterone produced by the enzymatic action of 5 α -reductase.

B. Sequence of events in sex differentiation

Sex differentiation proceeds in a cascadelike manner, with a series of temporally regulated successive steps at different levels of differentiation. After the primordial germ cells migrate into the undifferentiated gonads, early embryonic testes develop under the influence of

testis-determining factor (TDF) if a Y chromosome is present. TDF is identical with the Y-specific sequences of the SRY region (see p. 386). During normal male differentiation, the further development of the müllerian ducts is suppressed by the müllerian inhibitor factor. Testosterone can exert its effect only in the presence of an appropriate intracellular receptor (androgen receptor TFM, see p. 390).

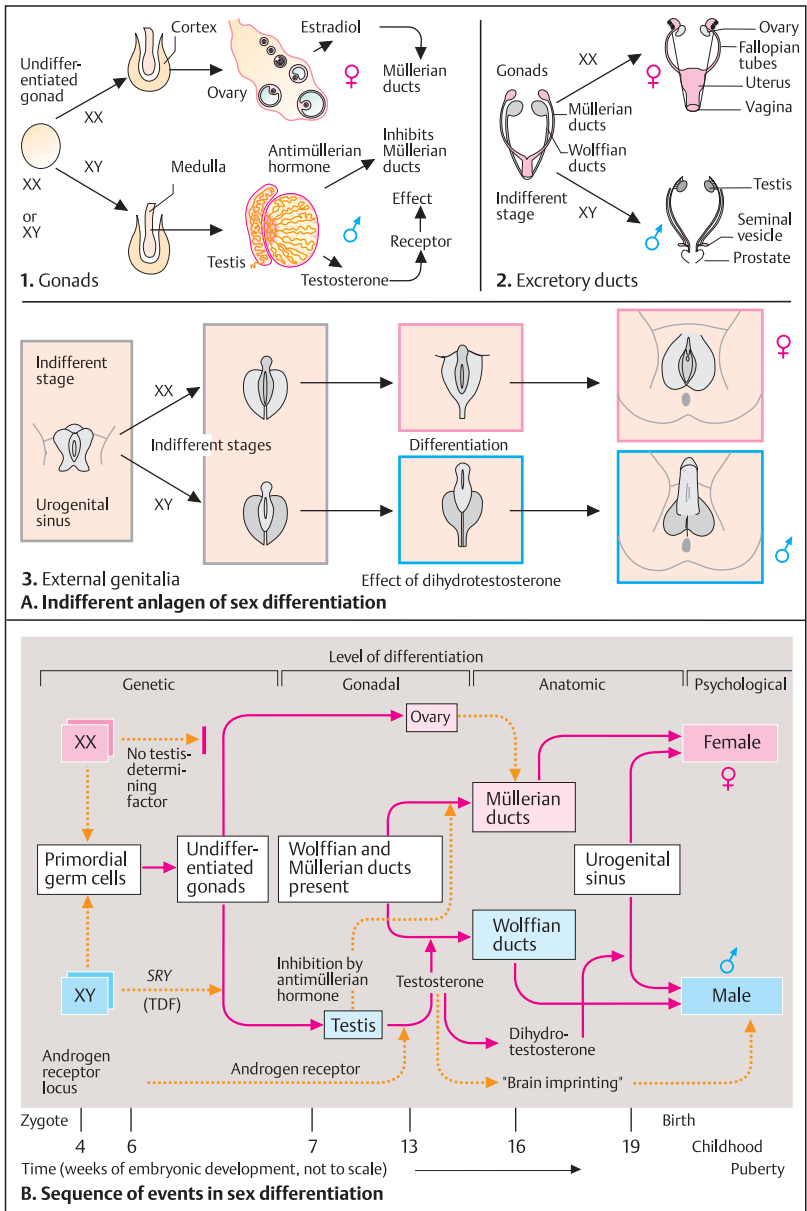
When a Y chromosome is not present or when the SRY region is missing or altered by mutation, testes are not formed. In this case the wolffian ducts cease to develop. In the absence of testes, ovaries develop from the undifferentiated gonads; the wolffian ducts degenerate; and the müllerian ducts differentiate into uterine tubes, uterus, and the upper vagina.

Testosterone also has an effect on the central nervous system ("brain imprinting"). It is assumed that this is required for the psychosexual orientation apparent later in life. When testosterone is absent or ineffective due to a receptor defect, gender orientation is female.

In the majority of genetically determined disorders of sexual differentiation, gonadal and genital sex do not correspond (pseudohermaphroditism). In true hermaphroditism, where the gonads consist of both testicular and ovarian tissues, male and female structures exist side by side.

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Disorders of Sexual Development

Normal sexual development is the result of numerous genes. Mutation or chromosomal rearrangements of any of these genes cause partial or total failure of sex differentiation. The classification of genetically determined disorders of sexual development takes the different developmental processes into account. Pinpointing the basic defect is a prerequisite for diagnosis and treatment.

A. Male-determining region *SRY* on the Y chromosome

Normally, the male-determining Y-specific DNA sequences (*SRY*) remain on the Y chromosome during the homologous pairing and crossing-over during meiosis. However, since the male-determining region *SRY* is located very close to the pseudoautosomal region (PAR), crossing-over in the PAR border region may result in a transfer of the *SRY* region to the X chromosome. This results in a male individual with an XX karyotype (XX male). Conversely, if the *SRY* region is missing from a Y chromosome, a female phenotype with XY chromosomes (XY female) results.

B. Point mutations in the *SRY* gene

The human *SRY* gene has a single exon and encodes a 204-amino-acid protein from a 1.1 kb transcript. The middle section of the *SRY* protein consists of 79 highly conserved amino acids with DNA-bending and DNA-binding capability, the HMG box (high mobility group protein). Complete or partial gonadal dysgenesis results from point mutations and deletions in the *SRY* gene, in particular the HMG box. (Figure adapted from Wolf et al., 1992; for an update of mutations see McElreavey and Fellous, 1999). Sex reversal also results from mutations in the *SOX9* gene on chromosome 17 at q24 in campomelic dysplasia. (Foster et al., 1994; Wagner et al., 1994).

C. Androgen receptor

The fetal testis produces testosterone, the hormone that induces male sexual differentiation. Testosterone is taken up by cells of the target tissues (wolffian ducts and urogenital sinus) (1). In the urogenital sinus, testosterone is converted into dihydrotestosterone (DHT) by the enzyme 5 α -reductase. Both testosterone and dihydrotestosterone bind to an intracellular receptor (androgen receptor). The activated hormone–receptor complex (TR* or DR*) acts as a transcription factor for genes that regulate the differentiation of the wolffian ducts and the urogenital sinus. Thus, normal male fetal development is dependent on normal biosynthesis of testosterone and normal receptors. Androgen receptor mutations lead to disorders of sexual development (2) with X-chromosomal inherited complete or incomplete androgen resistance (testicular feminization, TFM).

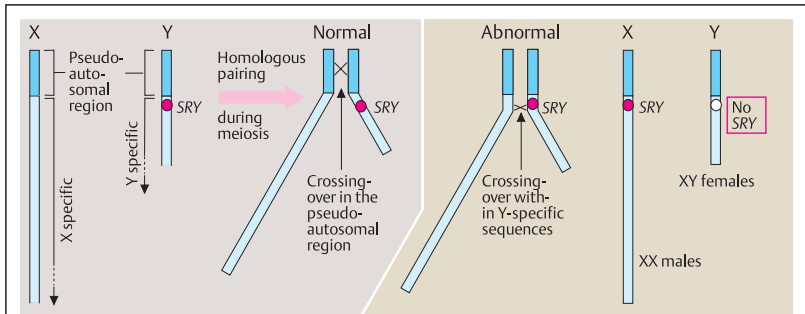
verted into dihydrotestosterone (DHT) by the enzyme 5 α -reductase. Both testosterone and dihydrotestosterone bind to an intracellular receptor (androgen receptor). The activated hormone–receptor complex (TR* or DR*) acts as a transcription factor for genes that regulate the differentiation of the wolffian ducts and the urogenital sinus. Thus, normal male fetal development is dependent on normal biosynthesis of testosterone and normal receptors. Androgen receptor mutations lead to disorders of sexual development (2) with X-chromosomal inherited complete or incomplete androgen resistance (testicular feminization, TFM).

Classification of genetically determined disorders of sexual development

1. Defects of sex determination due to mutation or structural aberration of the *SRY* region on the Y chromosome (e.g., XY gonadal dysgenesis, XX males, and others)
2. Defects of androgen biosynthesis (e.g., adrenogenital syndrome due to 21-hydroxylase deficiency, see p. 392)
3. Defects of androgen receptors (testicular feminization)
4. Defects of the müllerian inhibition substance (so-called hernia uteri syndrome)
5. XO/XY gonadal dysgenesis

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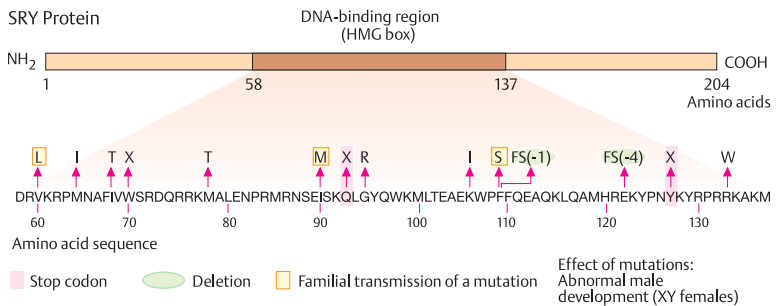
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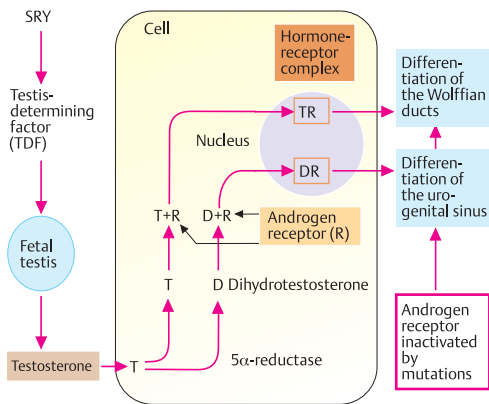
1. SRY remains on the Y chromosome

2. Transfer of SRY to the X chromosome

A. Male-determining region SRY on the Y chromosome



B. Point mutations in the SRY gene



1. Scheme of functional events

C. Androgen receptor and TFM



2. Phenotype

Congenital Adrenal Hyperplasia

This disorder, also called adrenogenital syndrome (AGS, McKusick 201910), is caused by a genetically determined deficiency of cortisol, a steroid hormone produced in the fetal adrenal cortex. A compensatory increase in adrenocortical hormone (ACTH) excretion leads to secondary enlargement (hyperplasia) of the adrenal cortex (congenital adrenal hyperplasia), increased production of prenatal steroids and their metabolites with androgenic effects, and incomplete female sex differentiation.

A. Clinical phenotype and genetics

Girls are born with ambiguous or virilized genitalia (1). The adrenal cortex is enlarged (2). Increased production of androgenic metabolites causes masculinization. The cortisol deficiency (3) leads to life-threatening crises due to loss of sodium chloride (salt-wasting) that require prompt treatment. AGS is an autosomal recessive heritable disorder (4). Untreated girls develop a male physical appearance (5). In boys, the early signs are limited to salt-wasting. Initially, skeletal maturation is accelerated and the children are tall for their age; however, they stop growing prematurely and eventually are too short. Besides the classic form of the disorder with a frequency of 1:5000, there are other forms with less pronounced masculinization due to different mutations.

B. Biochemical defect

The enzymatic conversion of progesterone to deoxycortisol (DOC) by hydroxylation at position 21 (steroid 21-hydroxylase) is decreased. As a result, the concentration of 17-hydroxyprogesterone is increased.

C. Gene locus and gene structure

21-Hydroxylase is encoded by the *CYP21* gene (formerly called *CYP21B* and *21-OHB*), a member of the cytochrome P450 oxidase gene family. This gene is located within the class III genes of the major histocompatibility complex on the short arm of human chromosome 6. It is part of a tandem paired arrangement of three other genes: active *C4A* and *C4B* genes and a 96–98% homologous inactive *CYP21P* gene, a pseudogene due to intragenic deletions resulting in stop codons (formerly called *CYP21A* or

21-OHA). These genes originated from a duplication event in evolution. The *CYP21* (*21-OHB*) gene consists of 10 exons spanning almost 6 kb of genomic DNA (the actual distance of 30 kb to the *C4A* and *C4B* genes is not shown to scale).

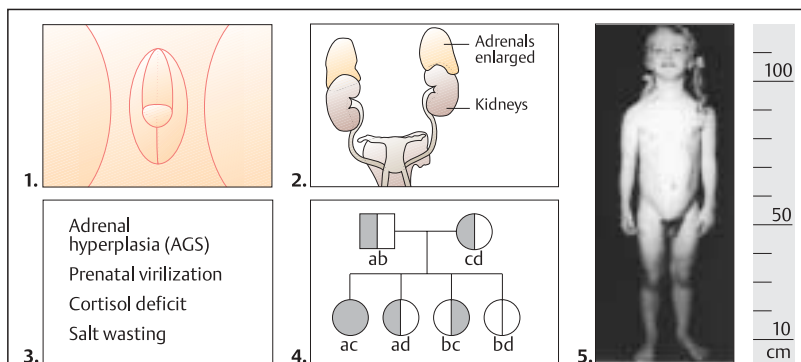
D. Molecular genetic analysis

Point mutations, deletions, and duplications occur in the *CYP21* gene. The deletions and duplications result from misalignment of the homologous chromatids during meiosis and unequal crossing-over. Deletions occur in about 20–25% of patients with classic 21-hydroxylase deficiency. Duplications have no clinical consequences. Deletions and duplications can be easily detected by Southern blot analysis. The most frequent type of deletion is loss of a 30 kb region including the 3' part of the *CYP21P* pseudogene, the entire *C4B* gene, and the 5' part of the *CYP21* gene. The resulting fusion gene of *CYP21P* and *CYP21* carries a *TaqI* restriction site in the 5' region of *CYP21P* that is not present in the *CYP21* gene. Therefore, the fusion gene has a characteristic 3.2 kb *TaqI* fragment. This distinguishes the rearrangement from the normal *CYP21* gene, which has a characteristic 3.7 kb fragment. In the example shown, the *CYP21* gene (*21-OHB*) is represented by a 3.7 kb DNA fragment, the pseudogene *CYP21P* (*21-OHA*) by a 3.2 kb fragment after *TaqI* digestion (1). Thus, the normal pattern is a 3.7 kb and a 3.2 kb fragment (2). Homozygous deletion of either of the genes may be apparent by lack of either of the two fragments (3, 4). A heterozygous deletion shows reduced intensity (5) and a duplication shows increased intensity (6). (Figure adapted from New et al., 1989).

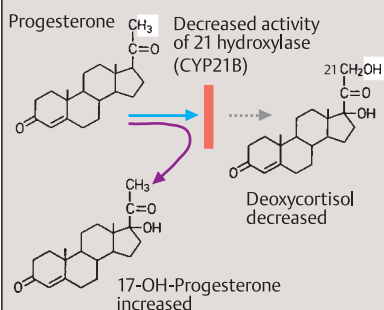
Another common mechanism for the origin of mutation in the *CYP21* gene is gene conversion. This involves nonreciprocal exchange between the closely linked *CYP21* and *CYP21P* genes, which results in transfer of mutations from the pseudogene *CYP21P* to *CYP21*.

References

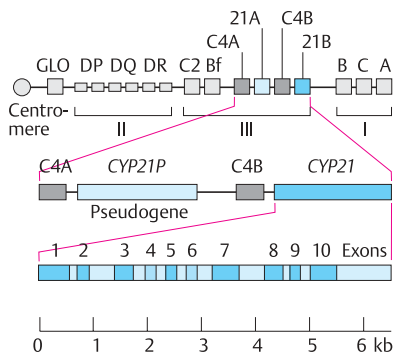
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A. Clinical phenotype and formal genetics

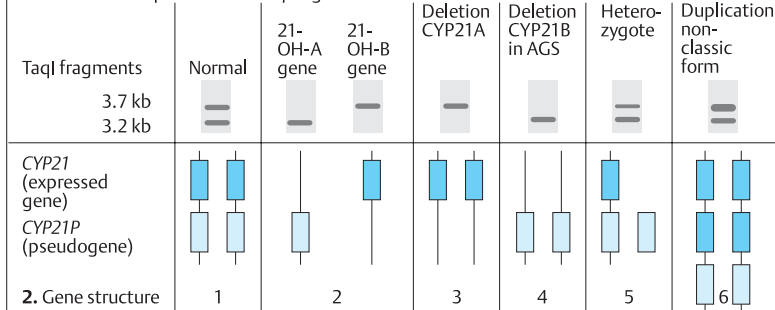


B. Biochemical defect



C. Gene locus and gene structure

1. Southern blot pattern after *TaqI* digestion



D. Molecular genetic analysis

Atypical Inheritance Pattern

Unstable Number of Trinucleotide Repeats

Heritable changes in the number of repeated groups of three nucleotides each (trinucleotide or triplet repeat) represent a new class of mutations in man for which there is no parallel in other organisms. They either occur within the gene and are translated or occur outside the gene in an untranslated region, and they are unstable during transmission through the germline. Unaffected persons may carry a premutation, which may be converted to a full mutation when passed through the germline to the next generation. Therefore, the effects of the mutation differ in severity in affected members within the same family. Occasionally, there is regression and a generation is skipped.

A. Genetic diseases with increased numbers of trinucleotides

Some important genetically determined diseases are based on a greater than normal number of trinucleotides: Huntington disease, fragile X syndrome, myotonic dystrophy, spinobulbar muscular atrophy type Kennedy, and spinocerebellar ataxia type 1. A total of 14 defined trinucleotide repeat diseases have been described (see Cummings and Zoghbi, 2000).

B. Huntington disease

Huntington disease is a progressive disease of the brain. Within 5–10 years, it leads to complete loss of motor control and intellectual abilities (1). It usually begins around age 40–50 with uncoordinated movements (chorea, St. Vitus' dance), excitation, hallucinations, and psychological changes. The disease is transmitted by autosomal dominant inheritance and shows complete penetrance. It presents an affected family with two difficult problems: (i) due to its late onset, carriers of the mutation have usually completed their family planning before the disease is manifest, and (ii) children of affected persons first learn as young adults that they are at a 50% risk of developing the disease later in life. Thus, the introduction of a direct predictive DNA diagnostic procedure is

very important. However, before such a genetic test is carried out, it must be established through genetic counseling that the persons at risk have decided for themselves whether they want to have the test performed. The gene is located on the distal short arm of chromosome 4 (2). It spans 210 kb and codes for a protein (called huntingtin) of important function. The 5' end of the gene contains numerous copies of a trinucleotide sequence consisting of cytosine, adenine, and guanine (CAG), a codon for the amino acid glutamine. Normally the gene has 10–34 CAG repeats; in patients there are 42–100. The diagnostic test (3) demonstrates that affected individuals (here, individuals 1, 2, and 4) have enlarged DNA fragments due to expanded CAG repeats. (Findings of the Institut für Humangenetik der Universität Göttingen with kind permission by Prof. W. Engel; Zühlke et al., *Hum. Mol. Genet.* 2:1467–1469, 1993).

C. Myotonic dystrophy (MDY1)

Myotonic dystrophy is an autosomal dominant hereditary disease that predominantly affects the central nervous and muscular systems (1). The myotonia causes a masklike facies (2). The disease is very variable and in many families shows increasing severity in consecutive generations (anticipation). An increased number of CTG repeats, more than 50 copies compared with 5–35 in normal individuals (3), is found immediately beyond the 3' end of the gene in affected persons. This is demonstrated in a Southern blot as an enlarged DNA fragment (4). (Schematic representation of a Southern blot at the gene locus D19S95, probe pBBO.7 after DNA cleavage with *EcoRI*. After Harley et al., *Lancet* 339:1125–1128, 1992).

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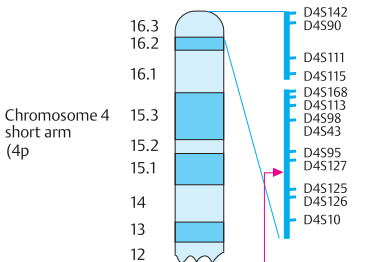
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Disease (Examples)	Gene	Frequency	Tri-nucleotide	Normal Number	Mutant Allele	Chromosome
Huntington disease	<i>HD</i>	1:10 000	(CAG) _n	0–26	36–121	4p16.3
Fragile X syndrome	<i>FMR1</i>	1:5 000	(CGG) _n	6–50	52–500	Xq27.3
Myotonic dystrophy	<i>DMPK</i>	1:8 000	(CTG) _n	5–37	50–500	19q13.2
Spinal-bulbar muscular atrophy (Kennedy)	<i>SBMA</i>	<1:50 000	(CAG) _n	11–31	36–65	Xq11–12

A. Genetic diseases due to increased number of trinucleotides

Severe progressive disease of the central nervous system
Loss of motor and intellectual control
Onset age 25–60
Autosomal dominant
CAG repeat size increased
Predictive diagnosis possible, but problematic

1. Main manifestations



2. Localization of the gene

Affected individuals 1, 2, and 4 have expanded CAG repeats

Expanded (CAG)_n repeats in Huntington disease (n = 36–121)

Normal (CAG)_n repeats (n = 10–26)

3. Diagnostic test

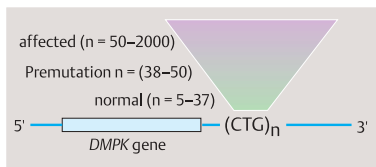
B. Huntington disease

Muscle weakness,
Myotonia, mask-like face,
Cataract, alopecia,
Variable expression,
Autosomal dominant,
CTG repeat increased



1. Main manifestations

2. Phenotype



3 Expanded CTG repeat in Myotonic Dystrophy

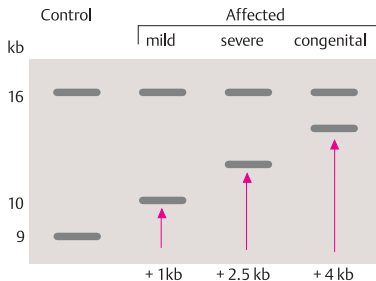


Diagram of a Southern blot at gene locus D19S95 (probe pBB0.7)

4. Correlation with degree of severity

C. Myotonic Dystrophy

Fragile X Syndrome

The fragile X syndrome (McKusick 309550; other designations: fraX syndrome, X-chromosomal mental retardation with fragile site on the X chromosome, Martin–Bell syndrome) is the most frequent form of hereditary mental retardation in males, with a frequency of about 1:2000–4000 individuals. The responsible mutation usually consists of an increased number of unstable trinucleotide repeats. Unlike in classic X-chromosomal inheritance, there are males without manifestations, and a large proportion of female carriers show partial manifestations.

The unstable expansion of a trinucleotide repeat (CGG) is located in the 5'-untranslated region of the *FMR1* gene. Recent findings indicate that an increase beyond 200 repeats impedes the migration of the 40S ribosomal subunit. This causes translational suppression.

A. Phenotype

The phenotype is very variable. The mental retardation varies; there is no distinct neurological dysfunction. In adult males, the testes are enlarged (macroorchidism). Affected individuals can usually be integrated well into the family and learn to function in a familiar environment.

B. Fragile site Xq27.3

The gene locus (FRAXA) for the gene (*FMR1*) is located on the distal long arm of the X chromosome in region 2, band 7.3 (Xq27.3). In this region the great majority of patients and some of the female heterozygotes show a constriction (fragile site) in the affected X chromosome in about 2–25% of metaphases. The constriction must be induced by folic acid deficiency in the culture medium, and it must be differentiated from other fragile sites in this region.

C. Expanded CGG repeats in the fragile X syndrome

The heritable unstable sequences explain two unusual characteristics of the fraX syndrome; (i) the transition from a premutation (about 60–200 CGG repeats) without clinical manifestation into a full mutation (more than 200 CGG repeats) during transmission through the germline, and (ii) differences in the FRAXA locus within a given family.

Fragile X syndrome is heritable as an X-linked dominant trait. The risk of transmission and clinical manifestation varies according to the type of mutation (premutation or full mutation), the gender of the patient and of the parent carrying an expanded trinucleotide repeat, and the relationship within the family. Males with the full mutation are mentally retarded and do not reproduce. Heterozygous females for the full mutation have a risk of variable mental retardation of 50%. They transmit the full mutation to 50% of their offspring. A premutation present in a male ("normal male transmitter") is transmitted to all daughters and none of the sons. Female carriers of the premutation or full mutation have a 50% risk of transmitting the mutant allele. The actual risk of manifest fragile X syndrome depends on the number of CGG repeats and varies between 10% (60–69 repeats) and 50% (more than 100 repeats) for sons (Gene Clinics at <http://www.geneclinics.org>).

The number of CGG repeats is variable within a family. In the pedigree shown (1), individuals II-3 and III-1 have more than 200 repeats and have fragile X syndrome. Individuals I-2, I-3, II-1, and III-3 are carriers of the premutation with 79–82 repeats. The normal number of CGG repeats is 6 to about 50, the premutation is defined by about 55 to 100, and the full disease-causing mutation by more than 200 repeats (2).

The different numbers of CGG repeats can be demonstrated in Southern blots as DNA fragments of different sizes (3). The normal gene is represented by a small DNA fragment (S). A premutation leads to slightly enlarged fragments. The full mutation is characterized by large fragments (L). With this procedure, a reliable diagnosis of the genotype is possible. (Photograph of a Southern blot: *HindIII* digestion and hybridization with probe Ox1.1; P. Steinbach, Ulm).

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A. Phenotype



b



c



d



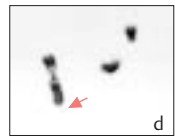
a



b

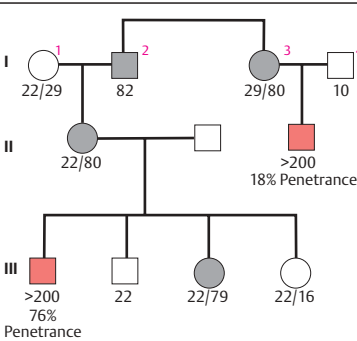


c



d

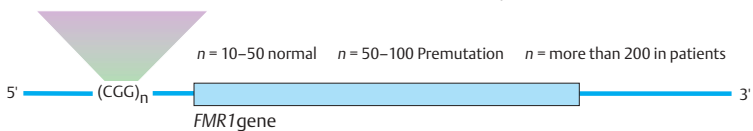
B. Fragile site Xq27.3



1. Variable number of CGG repeats

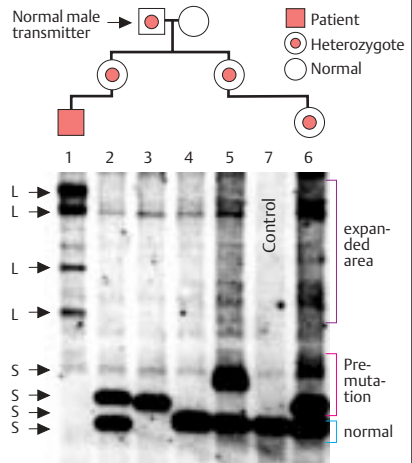
The number under the symbols correspond to the number of CGG trinucleotides of the FMR1 locus

1. Variable number of CGG repeats



2. Number of CGG repeats in mutation and premutation

C. Expanded CGG repeat in fragile X syndrome



3. Examination of a family with fraX syndrome

Imprinting Diseases

Prader–Willi syndrome (PWS) and Angelman syndrome (AS) are two distinct neurogenetic developmental disorders that result from different genetic lesions in a 3–4 Mb contiguous region of human chromosome 15 (15q11–13). This region is imprinted, i.e., genes on the maternal or the paternal allele only are expressed.

A. Two syndromes associated with the same chromosomal region

Prader–Willi syndrome is characterized by neonatal muscular weakness and feeding difficulties, followed in early childhood by reduced or lack of satiation control leading to massive obesity in many patients. Several other, variable features occur, such as mental retardation, characteristic facial features, short stature, hypopigmentation, behavioral problems, and other findings. In Angelman syndrome the developmental retardation is usually very severe. Nearly total lack of speech development, an abnormal electroencephalogram with tendency to seizures, and hyperactivity are almost always present.

B. Parental origin of the deletion

PWS results when the deleted chromosome involves the chromosome 15 of paternal origin (loss of one paternal allele 2 in the diagram of a Southern blot on the left). AS results when the deletion involves the chromosome 15 of maternal origin (loss of one maternal allele 1 in the scheme on the right).

C. Uniparental disomy

Uniparental disomy (UPD) is the presence of two chromosomes or genes from the same parent. The diagram of a Southern blot shows two different types of UPD in PWS: isodisomy and heterodisomy. In isodisomy the two parental alleles are identical (two maternal alleles 1 shown in the diagram). In heterodisomy the two alleles are of the same parental origin, but differ (one allele 1 and one allele 2, both of maternal origin, as shown).

D. Parent-of-origin deletion and uniparental disomy

A deletion and uniparental disomy have the same functional result, i.e., loss of the genetic

activity of one parental allele. The frequency of a deletion is about the same for PWS and AS (70% each), whereas the frequency of UPD differs considerably: 29% in PWS, 1% in AS.

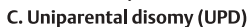
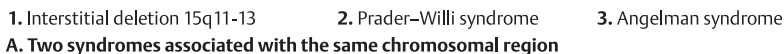
E. Chromosomal region 15q11–13 and imprinting center

Five genes known to date are transcribed from the paternal allele only, not from the maternal allele where they are constitutively repressed (blue squares). From the more distal gene *UBE3A* (a ubiquitin-protein ligase E3) only the maternal allele is transcribed. About 25% of cases of Angelman syndrome are caused by deletion or mutation of this gene. The breakpoints of the common large deletions occur predominantly in three breakpoint cluster regions. The imprinting center (IC) was originally defined by small deletions outside of the known imprinted genes. About 1% of patients with PWS and 4% with AS have imprinting center defects. The imprinted region on 15q11–13 shows a difference in methylation pattern between the maternal and the paternal allele. This is the basis of a diagnostic test.

(Data in E. kindly provided by Dr. Karin Buiting). Other imprinted chromosomal regions are also associated with human diseases, e.g., Beckwith–Wiedemann syndrome and some patients with Russell–Silver syndrome, among others.

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Karyotype – Phenotype Correlation

Autosomal Trisomies

A trisomy (the presence of three homologous chromosomes instead of the usual two) arises prezygotically during meiosis due to faulty distribution (nondisjunction) of a chromosome pair. It may also arise after fertilization (postzygotic) during somatic cell division (mitosis); in this case, trisomy is present in a certain proportion of cells (chromosomal mosaicism). Trisomy leads to a phenotype characteristic for the particular chromosome, although in humans most trisomies are lethal in early embryonic development.

A. Trisomy in jimsonweed (*Datura stramonium*)

In 1922, Blakeslee observed that triploid and tetraploid jimsonweed plants (*Datura stramonium*) differ little in phenotype. However, when plants contained three copies of only one of the 12 chromosomes (trisomy), and two each of the others, a characteristic appearance resulted for each of the trisomies (from Blakeslee, 1922).

B. Trisomies in the mouse

During the 1970s, A. Gropp and co-workers investigated the effect of trisomies on the development of the mouse. Trisomic mice, resulting from the segregation of translocations, had a developmental profile and certain morphological changes characteristic for each trisomy (1). Embryos with a chromosome missing (monosomies) died very early in gestation. (Figure from A. Gropp, 1982). A mouse embryo with trisomy 12 shows an open skull cap and other malformations on the 14th day of development (2), unlike other embryos of the same age (H. Winking, Lübeck, 1991; Boué et al., 1985). Only trisomy 19 is compatible with survival until birth (day 21), but the brain is too small (3). These animals die shortly after birth.

C. Autosomal trisomies in man

Of the 22 autosomes in man, only three occur regularly as trisomies in live-born infants: trisomy 21, trisomy 18, and trisomy 13. They differ in phenotype and course of disease. Other

trisomies are not observed in live-born infants because they are lethal in early embryonic life, and not compatible with life at birth (see p. 402). Trisomy 21 causes the clinical picture of Down syndrome (formerly called mongolism).

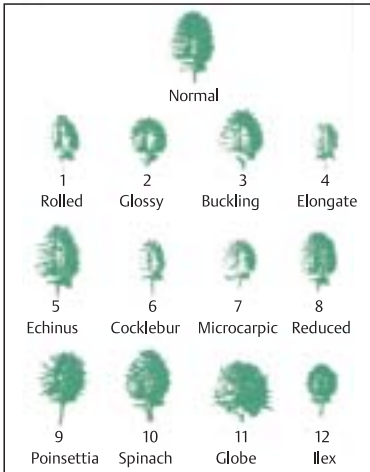
D. Nondisjunction as a cause of trisomy

Especially in trisomy 21, the frequency of nondisjunction depends on the age of the mother at the time of conception (1). The age of the father has very little or no influence.

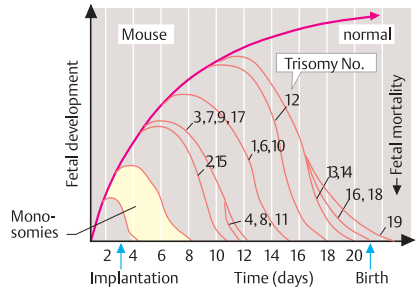
Nondisjunction may occur during the first or the second maturation division (meiosis I or meiosis II, p. 116) (2). The difference can be established by appropriate chromosomal markers. If nondisjunction occurs in meiosis I, the three chromosomes will be different ($1+1+1$), whereas if nondisjunction occurs during meiosis II, two of the three chromosomes will be identical ($2+1$). In humans, about 70% of nondisjunctions occur in meiosis I, and 30% in meiosis II.

References

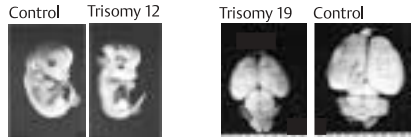
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A. Trisomy in jimson weed (*Datura stramonium*)



1. Developmental course of trisomies



2. Mouse embryo

3. Brain

B. Trisomies in the mouse



1. Trisomy 21

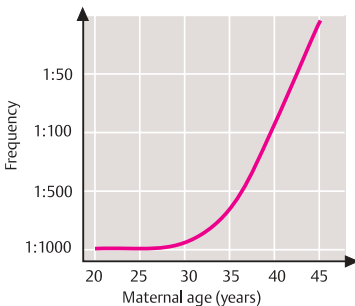


2. Trisomy 18



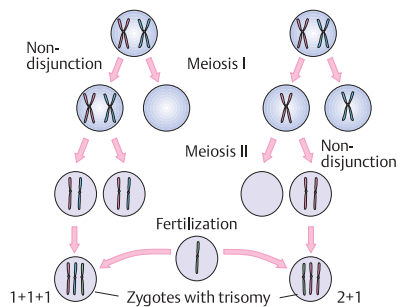
3. Trisomy 13

C. Trisomies in man



1. Trisomy 21 in liveborn infants

D. Nondisjunction as cause of trisomy



2. Faulty distribution of one chromosome

Other Numerical Chromosomal Deviations

In addition to the autosomal trisomies, there are other conditions associated with an abnormal number of chromosomes. They involve either the entire set of chromosomes (triploidy or tetraploidy) or the X chromosome or Y chromosome. Deviations from the normal number of X or Y chromosomes comprise about half of all chromosomal aberrations in man (total frequency about 1:400).

A. Triploidy

Triploidy is one of the most frequent chromosomal aberrations in man (1). Possible causes include a diploid spermatocyte, a diploid oocyte, or fertilization of an egg cell by two spermatozoa (dispermy, p.196). Triploidy usually leads to spontaneous miscarriage within the first four months of pregnancy. The fetus shows numerous severe malformations (2), such as cardiac defects, cleft lip and palate, skeletal defects, and others. The additional chromosome set may be of either maternal or paternal origin, with different clinical consequences.

B. Monosomy X (Turner syndrome)

Monosomy X (karyotype 45,XO) is a frequent chromosomal aberration, representing about 5% of those in humans at conception. However, of 40 zygotes with monosomy X, only one will develop to birth. The phenotypic spectrum is very wide. During the fetal stage, (1) lymphedema of the head and neck result in cystic hygroma, large multilocular thin-walled lymphatic cysts. Congenital cardiovascular defects, especially involving the aorta, and kidney malformations are frequent. An important component of the disease is the absence of ovaries, which develop only as connective tissue (streak gonads). Small stature is always a feature (average adult height about 150 cm). In newborns, webbing of the neck (pterygium colli) may be present as a residual of the lymphedema (clinical picture of Ullrich–Turner syndrome). On the other hand, the manifestations may be mild (2). Very frequently, pure monosomy is not present, but rather chromosomal mosaicism with normal cells (45,XO/46,XX) or a structurally altered X chromosome (deletion of the short arm, isochromosome of the long or short arm, ring chromosome, or other).

C. Additional X or Y chromosomes

An additional X chromosome in males (47,XXY) leads to the clinical picture of Klinefelter syndrome after puberty when untreated (1). This includes tall stature, absent or decreased development of male secondary sex characteristics, and infertility due to absent spermatogenesis. With an additional Y chromosome (47,XYY) no unusual phenotype results (2). Girls with three X chromosomes (47,XXX) are also physically unremarkable (3). However, learning disorders and delayed speech development have been observed in some of these children.

D. Wide spectrum of chromosomal aberrations in human fetuses

The relative proportions of the various trisomies observed in fetuses after spontaneous abortion differ. The most frequent is trisomy 16, which accounts for about 5% of all autosomal trisomies. Autosomal monosomies lead to death of the embryo within the first days or weeks. (Data after Lauritsen, 1982).

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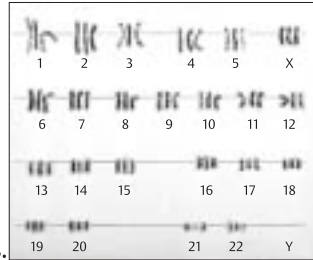
Triploidy

- Most frequent chromosomal aberration (15%) in fetuses following spontaneous abortion
- Severe growth retardation, early lethality
- Occasional liveborn infant with severe malformation
- Dispermia a frequent cause

1.

2.

3.

**A. Triploidy**

1.

2.

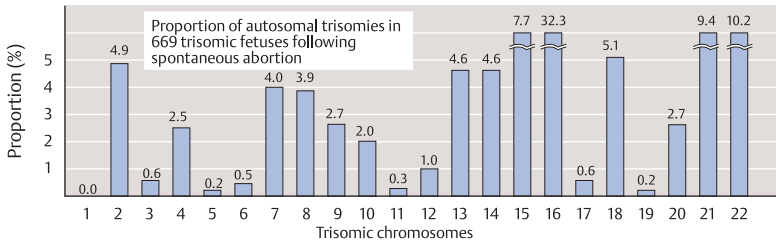
3.

**B. Monosomy X (Turner syndrome; 45, XO)**

1. XXY

2. XYY

3. XXX

**C. Additional X or Y chromosome****D. Wide spectrum of chromosomal aberrations in human fetuses**

Deletions and Duplications

Deletions and duplications are important structural aberrations of chromosomes. Deletion, which causes hemizygoty and functional haploinsufficiency for the loci involved, may occur *de novo* or be the result of the meiotic segregation of a parental balanced reciprocal translocation (see p. 198). Duplication of a chromosomal segment leads to partial trisomy, resulting in functional imbalance of the genes contained in the involved region.

A. Deletion 5p–: Cri-du-chat syndrome

In 1963, Lejeune and his co-workers in Paris described children with a partial deletion of the short arm of a chromosome 5 (5p–) and retarded mental and physical development. About 15% of the parents show a translocation of chromosome 5. In these cases, the risk of recurrence of the disorder is increased. Affected infants have prolonged, high-pitched crying resembling that of a kitten (cri-du-chat, cat cry).

B. Deletion 4p–: Wolf–Hirschhorn syndrome

Described in 1964 independently by U. Wolf in Freiburg and K. Hirschhorn in New York and their co-workers, this is a characteristic phenotype resulting from a partial deletion of chromosomal material of the short arm of a chromosome 4. Variable but considerable mental and statomotoric retardation is associated with characteristic facial features (1, 2) and with midline defects (cleft palate, hypospadias), coloboma of the iris, congenital heart defects, and other malformations. In some patients the deletion can only be detected by FISH. The simplified map of 4p16 (3) shows the critical chromosomal region (WHSCR, Wolf–Hirschhorn critical region). (Figure adapted from Wright et al., 1999).

C. Microdeletion syndromes

Of the more than 20 different microdeletion syndromes (for review see Spinner and Emanuel, 1996; Budarf and Emanuel, 1997) three are presented here. The Williams–Beuren syndrome (1, McKusick 194050, 130160) usually presents with characteristic facial features (“elfinlike”), infantile hypercalcemia, supraaortic stenosis, growth retardation, and impaired mental development. The underlying deletion involves the long arm of chromo-

some 7 at q11.23. The gene for elastin (*ELN*) is lost most frequently. Deletion of 22q11 leads to a group of clinically different but overlapping disorders (DiGeorge syndrome, McKusick 188400), characterized by absence or hypoplasia of the thymus and the parathyroid glands and malformations of the aortic arch; velocardiofacial syndrome, McKusick 192430; conotruncal cardiac defects, McKusick 217095; and others (2). The Rubinstein–Taybi syndrome (McKusick 180849) is characterized by typical facial features (3), broad thumbs and toes, and their associated radiological changes, mental retardation, and other features. A deletion of 16p13.3 is detectable in about 12% of patients. Point mutations in the CREB-binding gene (*CBP* gene) cause this disorder.

D. Phenotype of duplication 5q at different ages

A unique duplication illustrates the similar facial phenotypes at different ages: in a fetus at 22 weeks gestation (1), in a 5-month-old infant (2), and in an 8-year-old child. The affected individuals are siblings in one family. The partial duplication 5q33–qter resulted from a paternal reciprocal translocation (Passarge et al., 1982). A number of other duplications are associated with characteristic phenotypes.

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7 days



9 months



3 years



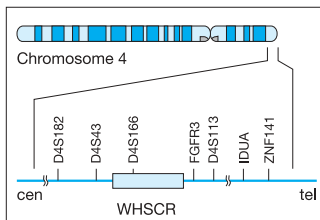
6 years

A. Deletion 5p--: Cri-du-chat syndrome

1. Age: 1 1/4 years



2. Age: 4 years



3. Scheme of physical map of 4p16

B. Deletion 4p--: Wolf-Hirschhorn syndrome

1. Williams-Beuren



2. Del22q11



3. Rubinstein-Taybi syndrome

**C. Other microdeletion syndromes (examples)**

1. Fetus: 22nd week



2. 5 months



3. 8 years

D. Phenotype of duplication 5q at different ages

A Brief Guide to Genetic Diagnosis

Principles

The diagnosis of a genetic disease requires a systematic approach that takes clinical and genetic considerations into account. Whereas clinical medicine tends to classify diseases according to organ system, age of onset, gender, or primary method of detection (radiology, imaging techniques), medical genetics, like pathology, is oriented towards the basic cause or lesion, in this case the gene or genes affected by a relevant genetic change. Genetic diagnosis is based on an interdisciplinary analysis of all clinical and laboratory data from a genetic perspective.

A. Genetic diagnosis, a multistep procedure

The phenotype, which is the clinical manifestations including individual and family history in medical terms, is the starting point. The first decision the medical geneticist must make is whether a pattern of manifestations can be recognized. Tools that can assist in this decision include training and personal experience; appropriate textbooks and other literature; and online search systems such as OMIM, MEDLINE, PubMed, POSSUM, London Dysmorphology Data Base for congenital malformations, and cytogenetic databases.

If a disease pattern can be recognized, the next decision concerns the category of disease. Although difficult to establish in practice, the disease category is important for the next steps to be taken. For this purpose, the McKusick catalog of human genes and diseases, Mendelian Inheritance in Man (MIM) and its online version (OMIM) are indispensable. The possibility of genetic heterogeneity must be considered at this stage. The term genetic heterogeneity refers to a phenotype (disease) that has different causes. A particular phenotype may be caused by mutations at different loci (*locus heterogeneity*) or by different mutant alleles at the same locus (*allele heterogeneity*). All genetic diagnostic procedures should be preceded by genetic counseling, which properly includes obtaining the (informed) consent of the persons involved.

B. Genotype analysis by PCR typing of a polymorphic restriction site

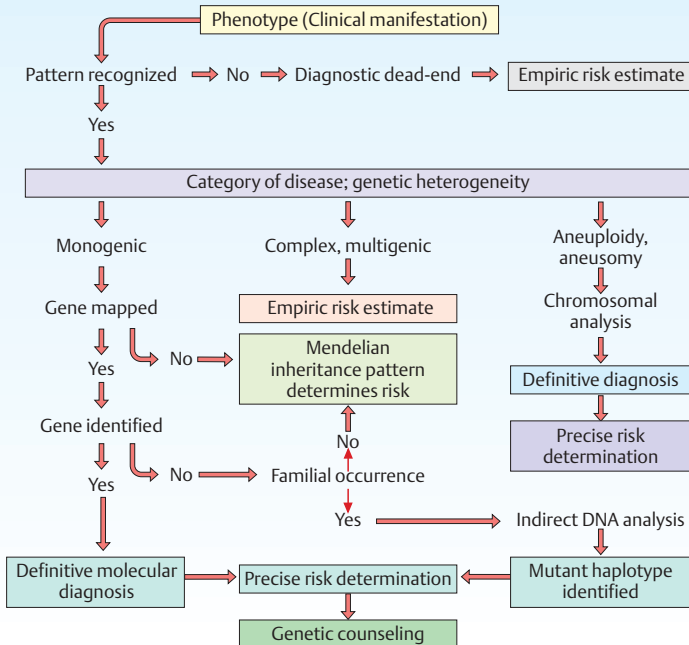
Genotype analysis by PCR typing of a polymorphic restriction site is preferred to the more laborious Southern blot hybridization (see p. 62). (Figure adapted from Strachan and Read, 1999).

C. Protein truncation test (PTT)

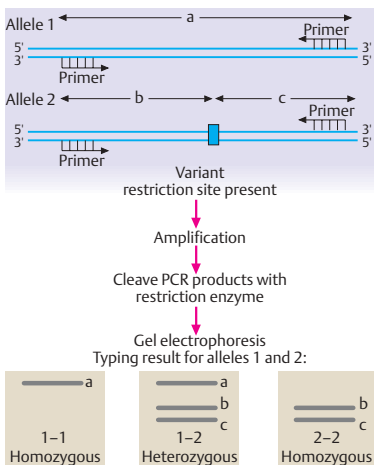
This is a test for frameshift, splice, or nonsense mutations that leads to a truncated protein due to an early stop codon created downstream of the mutation. The truncated protein is detected in an assay based on an in-vitro translation system. The translation will be interrupted at a premature stop codon resulting from the mutation. The size of the newly translated protein is determined by gel electrophoresis. PTT detects the approximate location of the mutation as reflected by the size of the mutant protein. PTT is useful in studying genes with frequent nonsense mutations, such as the *APC*, *BRCA1*, and *BRCA2* genes. However, it cannot be applied for genes with frequent missense mutations. The figure shown here is highly schematic. (Adapted from Strachan and Read, 1999; and Beaudet, 1998).

References

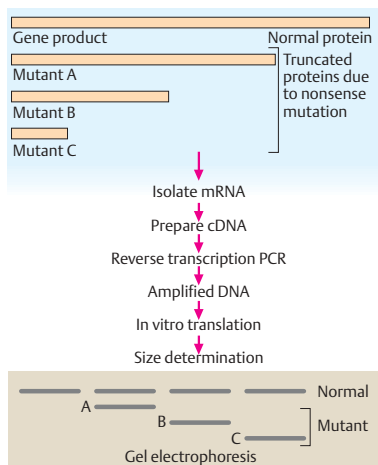
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A. Genetic diagnosis, a multistep procedure



B. Genotype analysis by PCR typing



C. Protein truncation test (scheme)

Detection of Mutations without Sequencing

In addition to the detection of mutations by different DNA fragments in Southern blots (p. 62), there are methods based on differences in the hybridization of mutated and normal segments of DNA. Incomplete hybridization is determined by using short segments of single-stranded DNA (oligonucleotides) with a sequence complementary to the investigated region (see A). Other methods are based on demonstrating incomplete hybridization with mRNA (see B) or on the fact that a hybridized segment of normal and mutant DNA is less stable than normal DNA.

A. Detection of a point mutation by oligonucleotides

Short segments of DNA (oligonucleotides) are used to determine whether there is a mutation in a segment of DNA (1, normal DNA; 2, mutation from G to A). An oligonucleotide is a synthetically produced DNA segment about 20 nucleotides long; its sequence is complementary to a corresponding segment of the investigated gene. It hybridizes completely with its complementary segment (3). If a mutation, here from G to A (1), is located in this region, hybridization will not be perfect at this site (mismatch) (4). On the other hand, an oligonucleotide that is complementary to the DNA segment with the mutation will hybridize completely (allele-specific oligonucleotide, ASO) (5). This hybridizes incompletely with the normal DNA (6). By parallel use of both nucleotides, mutant and nonmutant DNA can be differentiated. The test results (7) show the hybridization of mutated DNA and of control DNA with the allele-specific oligonucleotides (ASO 1 for the control, ASO 2 for the mutation). Hybridization is indicated by a signal (dot-blot analysis).

B. Demonstration of a point mutation by ribonuclease A cleavage

The basis for this method is that a normal DNA strand hybridizes completely with mRNA from that region. Completely hybridized DNA and mRNA are protected from the effects of the RNA-splitting enzyme ribonuclease A (ribonuclease protection assay). Hybridization is in-

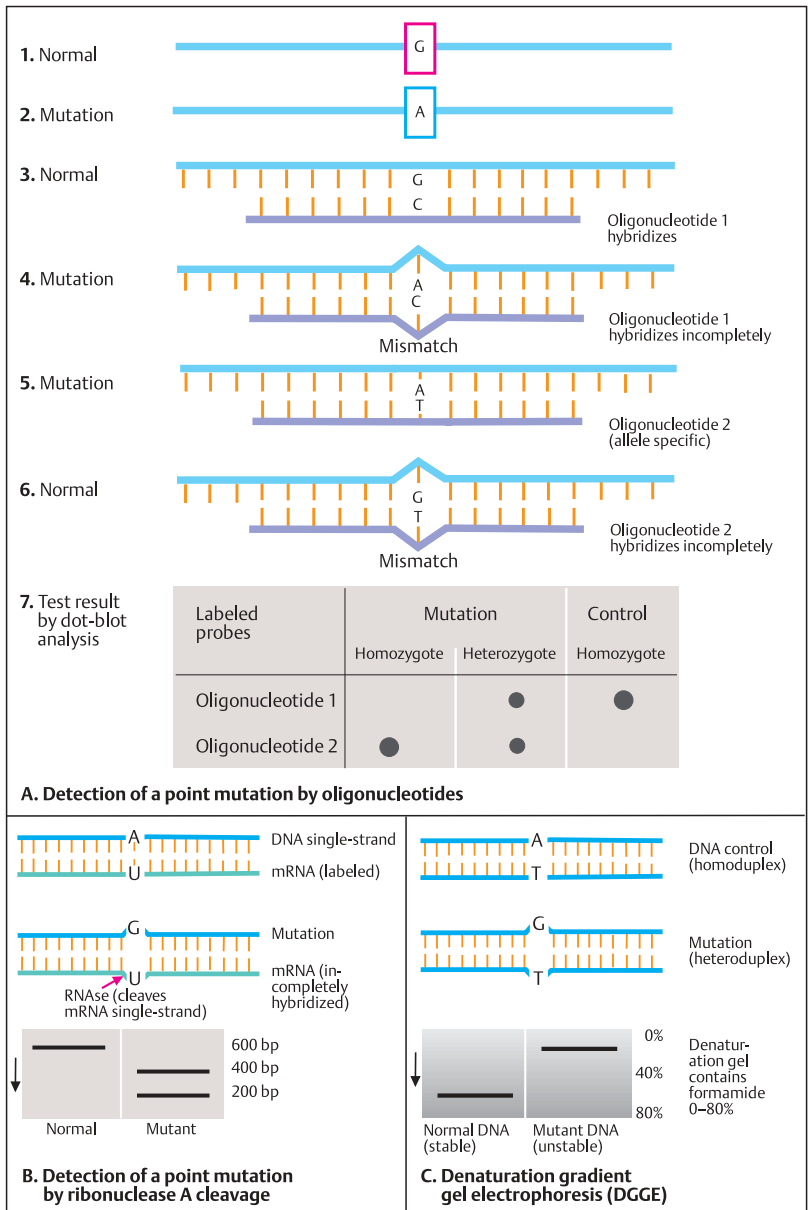
complete in the area of a mutation. In this region, mRNA will be cleaved by ribonuclease A (RNAase A). This can be demonstrated by Southern blot. There will be two fragments formed that together correspond to the size of the completely hybridized fragment (600 base pairs (bp), versus 400 and 200 bp).

C. Denaturing gradient gel electrophoresis

This method exploits differences in the stability of DNA segments with and without mutation. While double-stranded DNA of a control person is completely complementary (homoduplex), a mutation leads to a mismatch at the site of mutation (heteroduplex). This DNA is less stable than completely complementary DNA strands (it has a lower melting point). If normal DNA (control) and DNA with the mutation are placed in a gel with an increasing concentration gradient of formamide (denaturing gradient gel), the mutant and normal DNA can subsequently be differentiated in a Southern blot. The normal DNA remains stable to higher concentrations of formamide and migrates farther than mutant DNA, which dissociates earlier and therefore does not migrate as far.

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Chromosomal Location of Monogenic Diseases

Chromosomal Location of Human Genetic Diseases

Nowhere is the growth of knowledge about disease-causing mutations in human genes more apparent than in the mapping of their locations to specific sites on individual chromosomes. This progress is documented in twelve published editions of Mendelian Inheritance in Man. A Catalog of Human Genes and Disorders (MIM) by Victor A. McKusick, M.D., at the Johns Hopkins University School of Medicine. Initiated in the early 1960s, its first edition in 1966 contained a total of 1487 entries without a single autosomal gene mapped. This was achieved in 1968 at the time of the second edition (1545 entries). The subsequent editions reveal an entry-doubling time of about 15 years (3368 in the 6th edition (1983), 5710 entries in the 10th edition (1992), 8587 entries in the 12th edition (1998) and 10848 on 21 September 1999 (Hamosh et al., 2000). Since 1987 the McKusick catalog is internationally available online from the National Library of Medicine (Online Mendelian Inheritance in Man, OMIM, see references).

Regularly updated, OMIM has become a major source of information on human genes and genetic diseases. Each entry has a unique 6-digit identifying number and is assigned to one of five catalogs according to genetic category: (1) autosomal dominant, (2) autosomal recessive, (3) X-chromosomal, (4) Y-chromosomal, and (5) mitochondrial. Autosomal entries initiated since 1994 begin with the digit 6. The McKusick catalog has provided a systematic basis for the genetics of man comparable to the first periodic table of chemical elements by Dimitrij I. Mendelyev in 1869 or to the "Chronologisch-thematisches Verzeichnis sämtlicher Tonwerke Wolfgang Amade Mozarts" by Ludwig Alois Ferdinand Köchel in 1862.

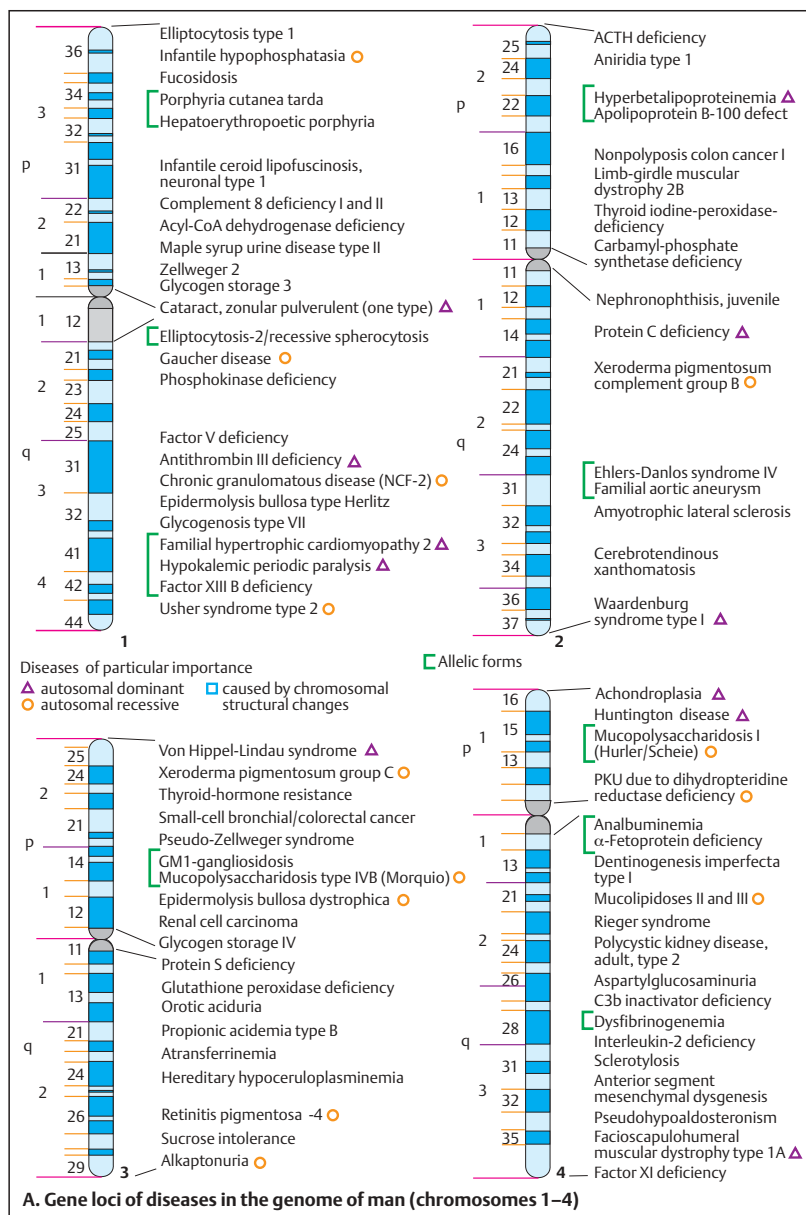
A special feature of the McKusick catalog is a map of disease-related gene loci assigned to specific chromosomal sites, called The Morbid Anatomy of the Human Genome. Since this first appeared in 1971 (3rd edition) on a single page,

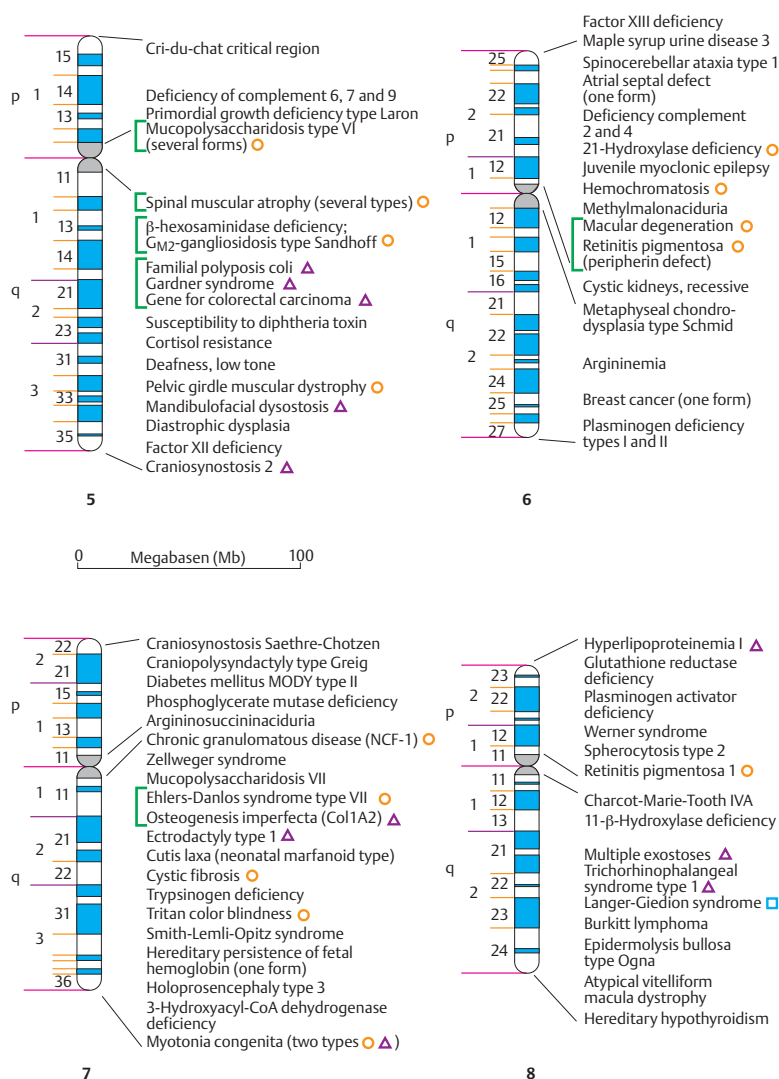
the complete information can no longer be presented in a readable printed version. The map of disease loci presented here on the next five pages, therefore, represents selected entries. For complete information, the reader is referred to the network of data available through OMIM. However, the maps shown on the following pages do provide an overview. More than 1000 disease genes have been mapped as of early 2000.

The McKusick catalog also reflects an important difference between customary clinical medicine and medical genetics. Whereas medicine classifies diseases according to their main manifestations, organ systems, age, gender, and other criteria related to the phenotype, medical genetics focuses on the genotype. The gene locus involved, the type of mutation, and genetic heterogeneity provide the basis for disease classification. This expands the concept of disease beyond the clinical manifestation and age of onset (see Childs, 1999).

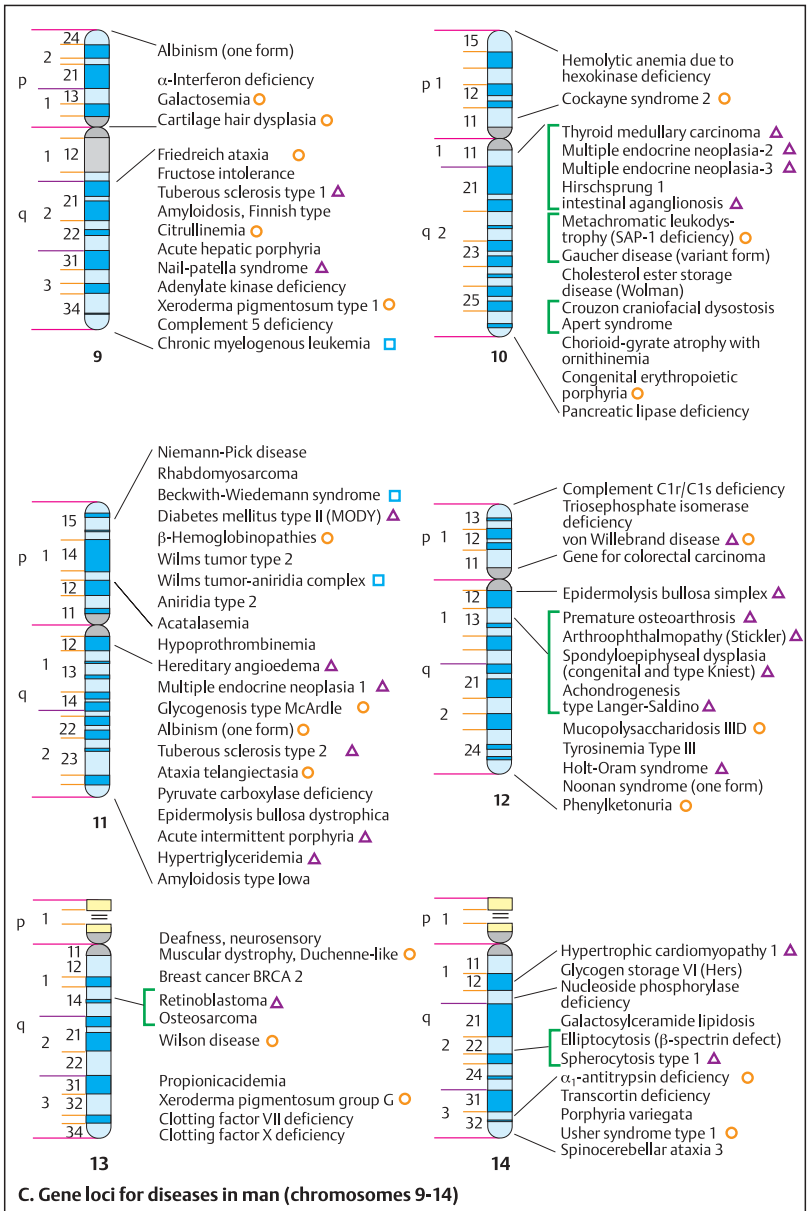
References

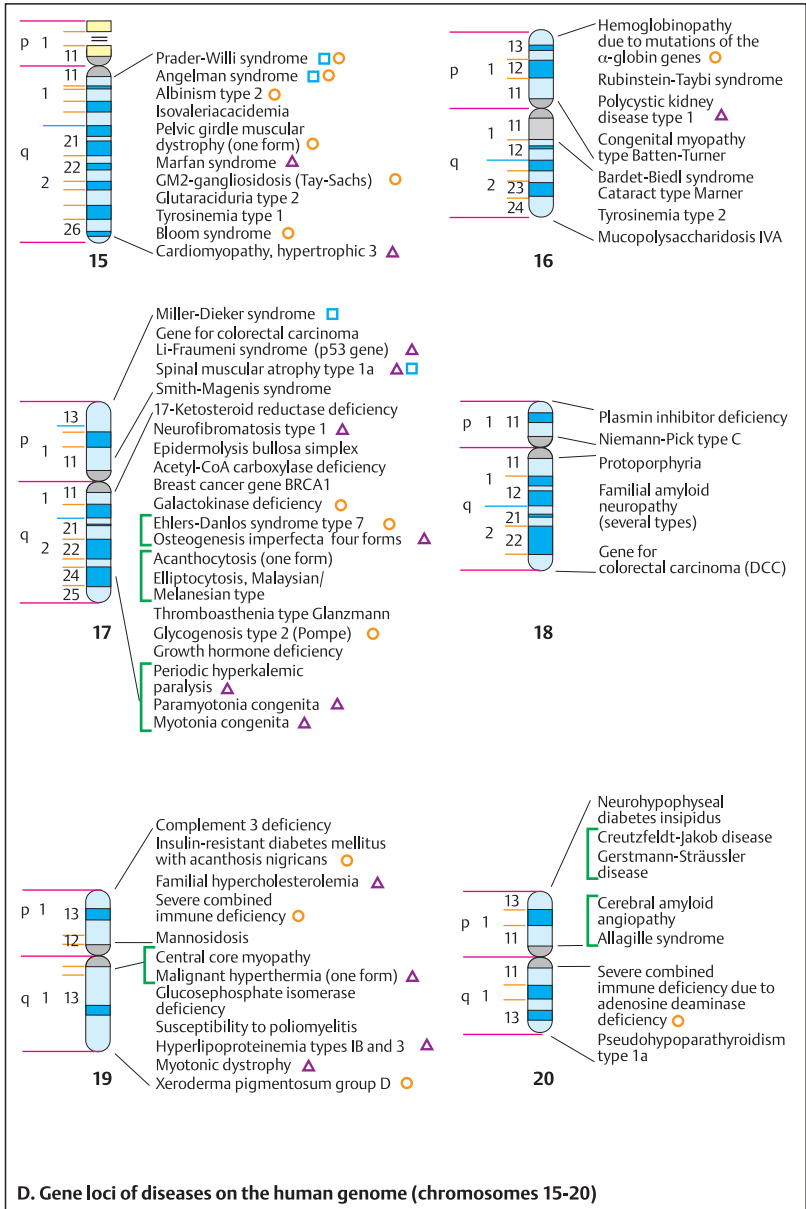
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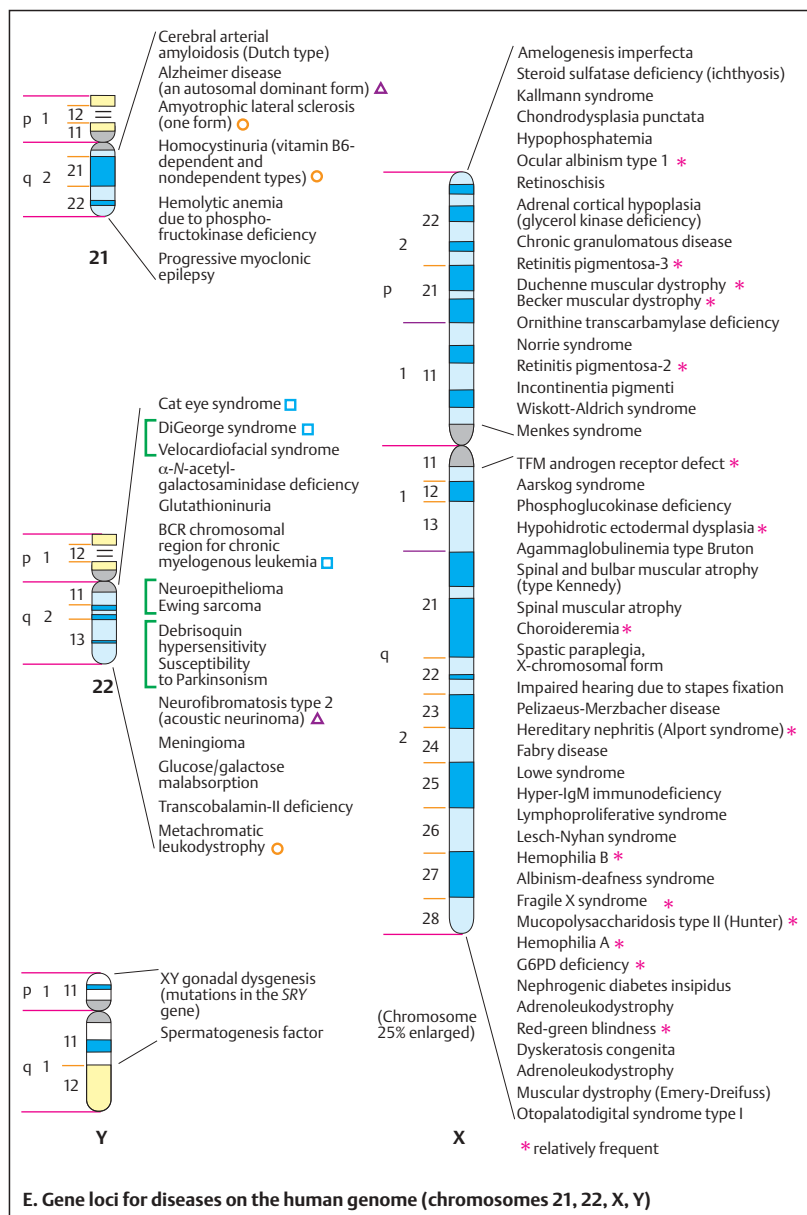




B. Gene loci of diseases on the human genome (chromosomes 5–8)







Chromosomal Location of Human Genetic Diseases

(Alphabetic list to the maps on pp. 410–415)

- Aarskog syndrome (X ch.)
- Acanthocytosis (one form) (ch. 17)
- Acatalasemia (ch. 11)
- Acetyl-CoA carboxylase deficiency (ch. 17)
- Achondroplasia (ch. 4)
- ACTH deficiency (ch. 2)
- Acute hepatic porphyria (ch. 9)
- Acute intermittent porphyria (ch. 11)
- Acyl-CoA dehydrogenase deficiency (ch. 1)
- Adenylate kinase deficiency (ch. 9)
- Adrenal cortical hypoplasia with glycerol kinase deficiency (X ch.)
- Adrenoleukodystrophy (X ch.)
- α -Interferon deficiency (ch. 9)
- Agammaglobulinemia (X ch.)
- Alagille syndrome (ch. 20)
- Albinism (one form) (ch. 9)
- Albinism (one form) (ch. 11)
- Albinism type 2 (ch. 15)
- Albinism – deafness syndrome (X ch.)
- Alkaptonuria (ch. 3)
- Alpha-N-acetylgalactosaminidase deficiency (ch. 22)
- Alpha-1-antitrypsin deficiency (ch. 14)
- Alpha-fetoprotein deficiency (ch. 4)
- Alzheimer disease (one form) (ch. 21)
- Amelogenesis imperfecta (X ch.)
- Amyloidosis, Finnish type (ch. 9)
- Amyloidosis type Iowa (ch. 11)
- Amyotrophic lateral sclerosis (one form) (ch. 21)
- Amyotrophic lateral sclerosis, juvenile (ch. 2)
- α -N-acetylgalactosaminidase deficiency (ch. 22)
- Analbuminemia (ch. 4)
- Androgen receptor defect (X ch.)
- Angelman syndrome (ch. 15)
- Aniridia type 1 (ch. 2)
- Aniridia type 2 (ch. 11)
- Anterior segmental mesenchymal dysgenesis (ch. 4)
- Antithrombin III deficiency (ch. 1)
- Apert syndrome (ch. 10)
- Apolipoprotein B-100 defect (ch. 2)
- Argininemia (ch. 6)
- Argininosuccinicaciduria (ch. 7)
- Arthroophthalmopathy (Stickler syndrome) (ch. 12)
- Aspartylglucosaminuria (ch. 4)
- Ataxia telangiectasia (one form) (ch. 11)
- Atransferrinemia (ch. 3)
- Atrial septal defect (one form) (ch. 6)
- Atypical vitelliform macular dystrophy (ch. 8)
- Bardet – Biedl syndrome (ch. 16)
- BCR chromosomal region for chronic myelogenous leukemia (ch. 22)
- Becker muscular dystrophy (X ch.)
- Beckwith – Wiedemann syndrome (ch. 11)
- Beta-hemoglobinopathies (ch. 11)
- Beta-hexosaminidase deficiency; GM²–gangliosidosis type Sandhoff (ch. 5)
- 11-Beta-hydroxylase deficiency (ch. 8)
- Bloom syndrome (ch. 15)
- Breast cancer gene BRCA1 (ch. 17)
- Breast cancer gene BRCA 2 (ch. 13)
- Breast cancer (one form) (ch. 6)
- Burkitt lymphoma (ch. 8)
- Carbamylphosphate synthetase – I deficiency (ch. 4)
- Cardiomyopathy, familial hypertrophic type 3 (ch. 15)
- Cartilage – hair dysplasia (ch. 9)
- Cat eye syndrome (ch. 22)
- C3b-inactivator deficiency (ch. 4)
- Central core myopathy (ch. 19)
- Cerebral amyloid angiopathy (ch. 20)
- Cerebral arterial amyloidosis (Dutch type) (ch. 21)
- Cerebrotendinosis xanthomatosis (ch. 2)
- Charcot – Marie – Tooth neuropathy type 1b (ch. 1)
- Charcot – Marie – Tooth neuropathy type II (ch. 1)
- Charcot – Marie – Tooth neuropathy type IVa (ch. 8)
- Cholesteryl ester storage disease (Wolman) (ch. 10)
- Chondrodysplasia punctata (X ch.)
- Choroid gyrate atrophy with ornithinemia (ch. 10)
- Choroideremia (X ch.)
- Chronic granulomatous disease (NCF-1) (ch. 7)
- Chronic granulomatous disease (NCF-2 deficiency) (ch. 1)
- Chronic granulomatous disease (X ch.)
- Chronic myelogenous leukemia (ch. 9)
- Citrullinemia (ch. 9)
- Clotting factor VII deficiency (ch. 13)
- Clotting factor X deficiency (ch. 13)

- Cockayne syndrome 2 (ch. 10)
 Colon cancer, familial nonpolyposis type 1 (ch. 2)
 Colorectal adenocarcinoma (ch. 12)
 Colorectal carcinoma (ch. 5 and 18)
 Colorectal carcinoma/Li–Fraumeni syndrome (ch. 17)
 Complement 2 and 4 deficiency (ch. 6)
 Complement 3 deficiency (ch. 19)
 Complement 5 deficiency (ch. 9)
 Complement 6, 7, and 9 deficiency (ch. 5)
 Complement 8 deficiency 1 and 2 (ch. 1)
 Complement C1r/C1s deficiency (ch. 12)
 Congenital erythropoietic porphyria (ch. 10)
 Congenital myopathy type Batten–Turner (ch. 16)
 Cortisol resistance (ch. 5)
 Craniopolysyndactyly type Greig (ch. 7)
 Craniosynostosis Saethre–Chotzen (ch. 7)
 Craniosynostosis type 2 (ch. 5)
 Creutzfeldt–Jakob disease (ch. 20)
 Cri du chat critical region (ch. 5)
 Crigler–Najjar syndrome (ch. 1)
 Crouzon craniofacial dysostosis (ch. 10)
 Cutis laxa (neonatal marfanoid type) (ch. 7)
 Cystic fibrosis (ch. 7)
 Deafness, low-tone (ch. 5)
 Deafness, neurosensory (ch. 13)
 Debrisoquin hypersensitivity (ch. 22)
 Dentinogenesis imperfecta type I (ch. 4)
 Diabetes mellitus type MODY (ch. 11)
 Diabetes mellitus, MODY type II (ch. 7)
 Diastrophic dysplasia (ch. 5)
 DiGeorge syndrome (ch. 22)
 Duchenne muscular dystrophy (X ch)
 Dysfibrinogenemia (ch. 4)
 Dyskeratosis congenita (X ch.)
 Ectrodactyly type 1 (ch. 7)
 Ehlers–Danlos syndrome type 4 (ch. 2)
 Ehlers–Danlos syndrome type 7 (ch. 7)
 Ehlers–Danlos syndrome type 7A1 (ch. 17)
 Elliptocytosis (β -spectrin defect) (ch. 14)
 Elliptocytosis, Malaysian/Melanesian type (ch. 17)
 Elliptocytosis-2/recessive spherocytosis (ch. 1)
 Epidermolysis bullosa dystrophica (ch. 3)
 Epidermolysis bullosa simplex (ch. 12)
 Epidermolysis bullosa simplex (ch. 17)
 Epidermolysis bullosa type Herlitz (ch. 1)
 Epidermolysis bullosa type Ogna (ch. 8)
 Ewing sarcoma (ch. 22)
 Fabry disease (X ch.)
 Facioscapulohumeral muscular dystrophy (ch. 4)
 Factor V deficiency (ch. 1)
 Factor XI deficiency (ch. 4)
 Factor XII deficiency (ch. 5)
 Factor XIIIa deficiency (ch. 6)
 Factor XIII B deficiency (ch. 1)
 Familial amyloid neuropathy (several types) (ch. 18)
 Familial aortic aneurysm (ch. 2)
 Familial hypercholesterolemia (ch. 19)
 Familial hypertrophic cardiomyopathy (ch. 1)
 Familial polyposis coli (ch. 5)
 Fragile X syndrome (X ch.)
 Friedrich's ataxia (ch. 9)
 Fructose intolerance (ch. 9)
 Fucosidosis (ch. 1)
 G6PD deficiency (X ch.)
 Galactokinase deficiency (ch. 17)
 Galactose epimerase deficiency (ch. 1)
 Galactosemia (ch. 9)
 Galactosylceramide lipidosis (ch. 14)
 Gardner syndrome (ch. 5)
 Gaucher disease (ch. 1)
 Gaucher disease (variant form) (ch. 10)
 Gerstmann–Sträussler disease (ch. 20)
 Glucose/galactose malabsorption (ch. 22)
 Glucosylphosphate isomerase deficiency (ch. 19)
 Glutaricaciduria type 2 (ch. 15)
 Glutathione peroxidase deficiency (ch. 3)
 Glutathione reductase deficiency (ch. 8)
 Glutathionuria (ch. 22)
 Glycogenosis type VII (ch. 1)
 Glycogenosis type McArdle (ch. 11)
 Glycogenosis type 2 (Pompe) (ch. 17)
 Glycogen storage type 3 (ch. 1)
 Glycogen storage type 4 (ch. 3)
 Glycogen storage VI (Hers) (ch. 14)
 GM1-gangliosidosis (ch. 3)
 GM2-gangliosidosis (Tay–Sachs) (ch. 15)
 GM2-gangliosidosis type Sandhoff (ch. 5)
 Growth hormone deficiency (ch. 17)
 Hemochromatosis (ch. 6)
 Hemoglobinopathies due to mutations of the α -globin genes (ch. 16)
 Hemolytic anemia due to hexokinase deficiency (ch. 10)
 Hemolytic anemia due to phosphofructokinase deficiency (ch. 21)
 Hemophilia A (X ch.)
 Hemophilia B (X ch.)
 Hepatoerythropoietic porphyria (ch. 1)
 Hereditary angioedema (ch. 11)
 Hereditary congenital hypothyroidism (ch. 8)

- Hereditary hypoceruloplasminemia (ch. 3)
 Hereditary nephritis (Alport syndrome) (X ch.)
 Hereditary persistence of fetal hemoglobin (one form) (ch. 7)
 Hirschsprung disease (chs. 10 and 13)
 Holoprosencephaly type 3 (ch. 7)
 Holt – Oram syndrome (ch. 12)
 Homocystinuria (B6-responsive and B6-nonresponsive forms) (ch. 21)
 Huntington disease (ch. 4)
 3-Hydroxyacyl-CoA dehydrogenase deficiency (ch. 7)
 21-Hydroxylase deficiency (ch. 6)
 Hyperbetalipoproteinemia (ch. 2)
 Hyper-IgM immune deficiency (X ch.)
 Hyperlipoproteinemia type 1 (ch. 8)
 Hyperlipoproteinemia type 1b (ch. 19)
 Hyperlipoproteinemia type 3 (ch. 19)
 Hypertriglyceridemia (ch. 11)
 Hypertrophic cardiomyopathy (ch. 14)
 Hypochondroplasia (ch. 4)
 Hypohidrotic ectodermal dysplasia (X ch.)
 Hypophosphatemia (X ch.)
 Hypoprothrombinemia (ch. 11)
 Immune deficiency, severe combined (ch. 19)
 Immunodeficiency due to ADA deficiency (ch. 20)
 Impaired hearing (lower frequencies) (ch. 5)
 Impaired hearing due to stapes fixation (X ch.)
 Infantile ceroid lipofuscinosis, neuronal type (ch. 1)
 Infantile hypophosphatasia (ch. 1)
 Insulin-resistant diabetes mellitus with acanthosis nigricans (ch. 19)
 Interleukin 2 deficiency (ch. 4)
 Intestinal aganglionosis (Hirschsprung) (ch. 10 and 13)
 Isovalericacidemia (ch. 15)
 Juvenile myoclonic epilepsy (ch. 6)
 Kallmann syndrome (X ch.)
 17-Ketosteroid reductase deficiency (ch. 17)
 Lamellar cataract (one type) (ch. 1)
 Langer – Giedion syndrome (ch. 8)
 Lesch – Nyhan syndrome (X ch.)
 Li – Fraumeni syndrome (ch. 17)
 Limb – girdle muscular dystrophy 2b (ch. 2)
 Lowe syndrome (X ch.)
 Lymphoproliferative syndrome (X ch.)
 Macular degeneration (ch. 6)
 Malignant hyperthermia (ch. 19, others)
 Mandibulofacial dysostosis (Franceschetti – Klein syndrome) (ch. 5)
 Mannosidosis (ch. 19)
 Maple syrup urine disease type 2 (ch. 1)
 Maple syrup urine disease type 3 (ch. 6)
 Marfan syndrome (ch. 15)
 Meningioma (ch. 22)
 Menkes syndrome (X ch.)
 Metachromatic leukodystrophy (ch. 22)
 Metachromatic leukodystrophy (SAP-1 deficiency) (ch. 10)
 Metaphyseal chondrodysplasia type Schmid (ch. 6)
 Methylmalonicaciduria (ch. 6)
 Miller – Dieker syndrome (ch. 17)
 Morquio syndrome B (ch. 3)
 Mucopolipidosis types II and III (ch. 4)
 Mucopolysaccharidosis type I (Hurler/Scheie) (ch. 4)
 Mucopolysaccharidosis type II (X ch.)
 Mucopolysaccharidosis type IVa (ch. 16)
 Mucopolysaccharidosis type IVb (ch. 3)
 Mucopolysaccharidosis type VI (Maroteaux – Lamy) (ch. 5)
 Mucopolysaccharidosis type VII (ch. 7)
 Multiple endocrine neoplasia type 1 (ch. 11)
 Multiple endocrine neoplasia type 2 (ch. 10)
 Multiple endocrine neoplasia type 3 (ch. 10)
 Multiple exostoses (ch. 8)
 Muscular dystrophy, Duchenne-like (ch. 13)
 Muscular dystrophy type Becker (X ch.)
 Muscular dystrophy type Duchenne (X ch.)
 Muscular dystrophy type Emery – Dreifuss (X ch.)
 Myotonia congenita (ch. 17)
 Myotonia congenita (two types) (ch. 7)
 Myotonic dystrophy (ch. 19)
 Myotubular myopathy (X ch.)
 Nail – patella syndrome (ch. 9)
 Nephrogenic diabetes insipidus (X ch.)
 Nephronophthisis, juvenile (ch. 2)
 Neuroepithelioma (ch. 22)
 Neurofibromatosis type 1 (ch. 17)
 Neurofibromatosis type 2 (acoustic neurinoma) (ch. 22)
 Neurohypophyseal diabetes insipidus (ch. 20)
 Niemann – Pick disease (ch. 11)
 Niemann – Pick type C (ch. 18)
 Noonan syndrome (one locus) (ch. 12)
 Norrie syndrome (X ch.)
 Nucleoside phosphorylase deficiency (ch. 14)
 Ocular albinism (X ch.)
 Ornithine transcarbamylase deficiency (X ch.)
 Oroticacidemia (ch. 3)
 Osteogenesis imperfecta (ch. 17)

- Osteogenesis imperfecta (COL1A2) (ch. 7)
 Osteosarcoma (ch. 13)
 Otopalatodigital syndrome type 1 (X ch.)
 Pancreatic lipase deficiency (ch. 10)
 Paramyotonia congenita (ch. 17)
 Pelizaeus – Merzbacher disease (X ch.)
 Pelvic girdle muscular dystrophy (ch. 5 and ch. 15)
 Periodic hyperkalemic paralysis (ch. 17)
 Phenylketonuria (PKU) (ch. 12)
 Phosphoglucokinase deficiency (X ch.)
 Phosphoglycerate mutase deficiency (ch. 7)
 Phosphokinase deficiency (ch. 1)
 PKU due to dihydropteridine reductase deficiency (ch. 4)
 Plasmin inhibitor deficiency (ch. 18)
 Plasminogen activator deficiency (ch. 8)
 Plasminogen deficiency, types I and II (ch. 6)
 Polycystic kidney disease (ch. 16 and 4)
 Porphyria cutanea tarda (ch. 1)
 Porphyria variegata (ch. 14)
 Prader – Willi syndrome (ch. 15)
 Primordial growth deficiency type Laron (ch. 5)
 Progressive myoclonic epilepsy (ch. 21)
 Propionicacidemia type A (ch. 13)
 Propionicacidemia type B (ch. 3)
 Protein C deficiency (ch. 2)
 Protein S deficiency (ch. 3)
 Protoporphyrin (ch. 18)
 Pseudoaldosteronism (ch. 4)
 Pseudohypoparathyroidism type 1a (ch. 20)
 Pseudo-Zellweger syndrome (ch. 3)
 Red – green blindness (X ch.)
 Renal cell carcinoma (ch. 3)
 Retinitis pigmentosa (peripherin defect) (ch. 6)
 Retinitis pigmentosa type 1 (ch. 8)
 Retinitis pigmentosa type 2 (X ch.)
 Retinitis pigmentosa type 3 (X ch.)
 Retinitis pigmentosa type 4 (ch. 3)
 Retinoblastoma (ch. 13)
 Retinoschisis (X ch.)
 Rhabdomyosarcoma (ch. 11)
 Rieger syndrome (ch. 4)
 Rubinstein – Taybi syndrome (ch. 16)
 Sclerolytosis (ch. 4)
 Sex reversal (XY females due to mutation in the SRY gene) (Y ch.)
 Small cell bronchial carcinoma/colorectal cancer (ch. 3)
 Smith – Lemli – Opitz syndrome (ch. 7)
 Smith – Magenis syndrome (ch. 17)
 Spastic paraplegia (X chromosomal form) (X ch.)
 Spermatogenesis factor (Y ch.)
 Spherocytosis type 1 (ch. 14)
 Spherocytosis type 2 (ch. 8)
 Spinal muscular atrophy (X ch.)
 Spinal muscular atrophy Ia (ch. 17)
 Spinal muscular atrophy IVa (ch. 8)
 Spinal muscular atrophy Werdnig – Hoffmann and other types (ch. 5)
 Spinocerebellar ataxia type 1 (ch. 6)
 Spinocerebellar ataxia type 3 (ch. 14)
 Spondyloepiphyseal dysplasia (congenital type) (ch. 12)
 Spondyloepiphyseal dysplasia (type Kniest) (ch. 12)
 Steroid sulfatase deficiency (ichthyosis) (X ch.)
 Sucrose intolerance (ch. 3)
 Susceptibility to diphtheria toxin (ch. 5)
 Susceptibility to Parkinsonism (ch. 22)
 Susceptibility to poliomyelitis (ch. 19)
 T cell leukemia/lymphoma (ch. 14)
 TFM androgen receptor defect (X ch.)
 Thrombasthenia type Glanzmann (ch. 17)
 Thyroid hormone resistance (ch. 3)
 Thyroid iodine peroxidase deficiency (ch. 2)
 Thyroid medullary carcinoma (ch. 10)
 Transcobalamin II deficiency (ch. 22)
 Transcortin deficiency (ch. 14)
 Trichorhinophalangeal syndrome type 1 (ch. 8)
 Triosephosphate isomerase deficiency (ch. 12)
 Tritan color blindness (ch. 7)
 Trypsinogen deficiency (ch. 7)
 Tuberous sclerosis type 1 (ch. 9)
 Tuberous sclerosis type 2 (ch. 11)
 Tuberous sclerosis type 3 (ch. 12)
 Tyrosinemia type 1 (ch. 14)
 Tyrosinemia type 2 (ch. 16)
 Usher syndrome type 1 (ch. 14)
 Usher syndrome type 2 (ch. 1)
 Velocardiofacial syndrome (ch. 22)
 Vitelline macular dystrophy (ch. 6)
 Von Hippel – Lindau syndrome (ch. 3)
 Von Willebrand disease (ch. 12)
 Waardenburg syndrome type 1 (ch. 2)
 Werner syndrome (ch. 8)
 Williams – Beuren syndrome (ch. 4)
 Wilms tumor – aniridia complex (ch. 11)
 Wilms tumor type 2 (ch. 11)
 Wilson's disease (ch. 13)
 Xeroderma pigmentosum complementation group B (ch. 2)
 Xeroderma pigmentosum group C (ch. 3)
 Xeroderma pigmentosum group D (ch. 19)
 Xeroderma pigmentosum group G (ch. 13)

Xeroderma pigmentosum type 1 (ch. 9)
XY gonadal dysgenesis (Y ch.)
Zellweger syndrome (ch. 7)
Zellweger syndrome type 2 (ch. 1)
Zonular cataract (ch. 1)

Caveat: There are numerous similar disorders caused by mutations of genes at other loci, sometimes with other modes of inheritance. This list and the corresponding maps are not complete, but give examples. For a complete map and list, see McKusick's catalog, 12th and OMIM edition, 1998.

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Glossary

Definitions of Genetic Terms

Acentric—refers to a chromosome or chromatid without a centromere.

Acrocentric (White, 1945)—refers to a chromosome with a centromere that lies very close to one of the ends, dividing the chromosome into a long and a very short arm.

Active site—the region of a protein that is mainly responsible for its functional activity.

Alkyl group—covalently joined carbon and hydrogen atoms as in a methyl or ethyl group.

Allele (Johannsen, 1909) or **allelomorph** (Bateson and Saunders, 1902)—one of several alternative forms of a gene at a given gene locus.

Alu sequences—a family of related DNA sequences, each about 300 base pairs long and containing the restriction site for the Alu restriction enzyme; about 500 000 copies of Alu sequences are dispersed throughout the human genome.

Amber codon—the stop codon UAG.

Ames test—a mutagenicity test carried out with a mixture of rat liver and mutant bacteria.

Aminoacyl tRNA—a transfer RNA carrying an amino acid.

Amplification—production of additional copies of DNA sequences.

Anaphase (Strasburger, 1884)—a stage of mitosis and of meiosis I and II. Characterized by the movement of homologous chromosomes (or sister chromatids) toward opposite poles of the cell division spindle.

Aneuploid (Täckholm, 1922)—having either fewer or more than the normal number of chromosomes.

Aneusomy—deviation from the normal presence of homologous chromosomal segments. Aneusomy by recombination refers to the duplication/deficiency resulting from crossing-over within an inversion (inverted region).

Annealing—to cause complementary single strands of nucleic acid to pair and form double-stranded molecules (DNA with DNA, RNA with RNA, or DNA with RNA).

Antibody—a protein (immunoglobulin) that recognizes and binds to an antigen as part of the immune response.

Antigen—a substance with a molecular surface structure that triggers an immune response, i.e., the production of antibodies, and/or that reacts with (its) specific antibodies (antigen-antibody reaction).

Antisense RNA—an RNA strand that is complementary to mRNA. Thus it cannot be used as a template for normal translation. The term antisense is generally used to refer to a sequence of DNA or RNA that is complementary to mRNA.

Apoptosis—programmed cell death, characterized by a series of regulated cellular events resulting in cell death, to eliminate a damaged cell or a normal cell no longer needed during development.

Attenuator—terminator sequence regulating the termination of transcription, involved in controlling the expression of some operons in bacteria.

Australopithecus—the genus of fossil Hominae from Eurasia. Walked erect; brain size between that of modern man and other modern primates; large and massive jaw. Lived about 4–5 million years ago.

Autonomously replicating sequence (ARS)—a DNA sequence that permits replication.

Autoradiography (Lacassagne and Lattes, 1924)—photographic detection of a radioactive substance incorporated by metabolism into cells or tissue. The distribution of the radioactively labeled substance can be demonstrated, e.g., in tissue, cells, or metaphase chromosomes by placing a photographic film or photographic emulsion in close contact with the preparation.

Autosome (Montgomery, 1906)—any chromosome except a sex chromosome (the latter usually designated X or Y). Autosomal refers to genes and chromosomal segments that are located on autosomes.

Auxotrophic (Ryan and Lederberg, 1946)—refers to cells or cell lines that cannot grow on minimal medium unless a certain nutritive substance is added (cf. prototrophic).

Backcross—cross of a heterozygous animal with one of its homozygous parents. In a double backcross, two heterozygous gene loci are involved.

Bacteriophage—a virus that infects bacteria. Usually abbreviated as phage.

Banding pattern (Painter, 1939)—staining pattern of a chromosome consisting of alternating light and dark transverse bands. Each chromosomal segment of homologous chromosomes shows the same specific banding pattern, characterized by the distribution and size of the bands, which can be used to identify that segment. The term was introduced by Painter in 1939 for the linear pattern of strongly and weakly staining bands in polytene chromosomes of certain diptera (mosquitoes, flies). Each band is defined relative to its neighboring bands. The sections between bands are interbands.

Barr body—see X chromatin.

Base pair (bp)—in DNA, two bases—one a purine, the other a pyrimidine—lying opposite each other and joined by hydrogen bonds. Normal base pairs are A with T and C with G in a DNA double helix. Other pairs can be formed in RNA.

B cells—B lymphocytes.

Bimodal distribution—refers to a frequency distribution curve with two peaks. If the frequency distribution curve of a population trait is bimodal, it is frequently evidence of two different phenotypes distinguished on a quantitative basis.

Bivalent (Haecker, 1892)—pairing configuration of two homologous chromosomes during the first meiotic division. As a rule the number of bivalents corresponds to half the normal number of chromosomes in diploid somatic cells. Bivalents are the cytogenetic prerequisite for crossing-over of nonsister chromatids. During meiosis, a trisomic cell forms a trivalent of the trisomic chromosomes.

Breakpoint—site of a break in a chromosomal alteration, e.g., translocation, inversion, or deletion.

CAT (or CAAT) box—a regulatory DNA sequence in the 5' region of eukaryotic genes; transcription factors bind to this sequence.

cDNA—complementary DNA synthesized by the enzyme reverse transcriptase from RNA as the template.

Cell cycle—life cycle of an individual cell. In dividing cells, the following four phases can be distinguished: G1 (interphase), S (DNA synthesis), G2, and mitosis. Cells that do not divide are said to be in the Go phase.

Cell hybrid—in culture, a somatic cell generated by fusion of two cells of different species. It contains the complete or incomplete chromosome complements of the parental cells. Cell hybrids are an important tool in gene mapping.

Centimorgan—a unit of length on a linkage map [100 centimorgans (cM) = 1 Morgan]. The distance between two gene loci in centimorgans corresponds to their recombination frequency expressed as percentage, i.e., one cM corresponds to one percent recombination frequency. Named after Thomas H. Morgan (1866–1945), who initiated the classic genetic experiments on *Drosophila* in 1910.

Centromere (Waldeyer, 1903)—chromosomal region to which the spindle fibers attach during mitosis or meiosis. It appears as a constriction at metaphase.

Chaperone—a protein needed to assemble or fold another protein correctly.

Chiasma (Janssens, 1909)—cytologically recognizable region of crossing-over in a bivalent. In some organisms the chiasmata move toward the end of the chromosomes (terminalization of the chiasmata) during late diplotene and diakinesis (see meiosis). The average number of chiasmata in autosomal bivalents is about 52 in human males, about 25–30 in females. The number of chiasmata in man was first determined in 1956 in a paper that confirmed the normal number of chromosomes in man (C. E. Ford and J. L. Hamerton, *Nature* 178 : 1020, 1956).

Chimera (Winkler, 1907)—an individual or tissue that consists of cells of different genotypes of prezygotic origin.

Chromatid (McClung, 1900)—longitudinal subunit of a chromosome resulting from chromosome replication; two chromatids are held together by the centromere and are visible during early prophase and metaphase of mitosis and

between diplotene and the second metaphase of meiosis. Sister chromatids arise from the same chromosome; nonsister chromatids are the chromatids of homologous chromosomes.

After division of the centromere in anaphase, the sister chromatids are referred to as daughter chromosomes. A *chromatid break* or a chromosomal aberration of the chromatid type affects only one of the two sister chromatids. It arises after the DNA replication cycle in the S phase (see cell cycle). A break that occurs before the S phase affects both chromatids and is called an isolocus aberration (*isochromatid break*).

Chromatin (Flemming, 1882)—the stained material that can be observed in interphase nuclei. It is composed of DNA, basic chromosomal proteins (histones), nonhistone chromosomal proteins, and small amounts of RNA.

Chromomere (Wilson, 1896)—each of the linearly arranged thickenings of the chromosome visible in meiotic and under some conditions also in mitotic prophase. Chromomeres are arranged in chromosome-specific patterns.

Chromosome (Waldeyer, 1888)—the gene-carrying structures, which are composed of chromatin and are visible during nuclear division as threadlike or rodlike bodies. *Polytene chromosomes* (Koltzoff, 1934; Bauer, 1935) are a special form of chromosomes in the salivary glands of some diptera larvae (mosquitoes, flies).

Chromosome walking—sequential isolation of overlapping DNA sequences in order to find a gene on the chromosome studied.

Cis-acting—refers to a regulatory DNA sequence located on the same chromosome (*cis*) as opposed to trans-acting on a different chromosome.

Cis/trans (Haldane, 1941)—in analogy to chemical isomerism, refers to the position of genes of double heterozygotes (heterozygotes at two neighboring gene loci) on homologous chromosomes. When two certain alleles, e.g., mutants, lie next to each other on the same chromosome, they are in the *cis* position. If they lie opposite each other on different homologous chromosomes, they are in the *trans* position. The *cis/trans* test (Lewis, 1951; Benzer, 1957) uses

genetic methods (*genetic complementation*) to determine whether two mutant genes are in the *cis* or in the *trans* position. With reference to genetic linkage, the expressions *cis* and *trans* are analogous to the terms coupling and repulsion (q.v.).

Cistron (Benzer 1957)—a functional unit of gene effect as represented by the *cis/trans* test. If the phenotype is mutant with alleles in the *cis* position and the alleles do not complement each other (*genetic complementation*), they are considered alleles of the same cistron. If they complement each other they are considered to be nonallelic. This definition of Benzer was later expanded (Fincham, 1959): accordingly, a cistron now refers to a segment of DNA that codes for a unit of gene product. Within a cistron, mutations in the *trans* position do not complement each other. Functionally, the term cistron can be equated with the term gene.

Clade—a group of organisms evolved from a common ancestor.

Clone (Webber, 1903)—a population of molecules, cells or organisms that have originated from a single cell or a single ancestor and are identical to it and to each other.

Cloning efficiency—a measure of the clonability of individual mammalian cells in culture.

Cloning vector—a plasmid, phage, or yeast artificial chromosome (YAC) used to carry a foreign DNA fragment for the purpose of cloning (producing multiple copies of the fragment).

Coding strand of DNA—the strand of DNA that bears the same sequence as the RNA strand (mRNA) that is used as a template for translation (sense RNA). The other strand of DNA, which directs synthesis of the mRNA, is the template strand (see antisense RNA).

Codominant—see dominant.

Codon (Crick, 1963)—a sequence of three nucleotides (a triplet) in DNA or RNA that codes for a certain amino acid or for the terminalization signal of an amino acid sequence.

Complementation, genetic (Fincham, 1966)—complementary effect of (restoration of normal function by) double mutants at different gene loci. An example, genetic complementation for xeroderma pigmentosum, is illustrated on page 82.

Concatemer—an association of DNA molecules with complementary ends linked head to tail and repeated in tandem. Formed during replication of some viral and phage genomes.

Concordance—the occurrence of a trait or a disease in both members of a pair of twins (mono- or dizygotic).

Conjugation—the transfer of DNA from one bacterium to another.

Consanguinity—blood relationship. Two or more individuals are referred to as consanguineous (related by blood) if they have one or more ancestors in common. A quantitative expression of consanguinity is the coefficient of inbreeding (q.v.).

Consensus sequence—a corresponding or identical DNA sequence in different genes or organisms.

Contig—a series of overlapping DNA fragments (contiguous sequences).

Cosmid—a plasmid carrying the cos site (q.v.) of λ phage in addition to sequences required for division. Serves as a cloning vector for DNA fragments up to 40 kb.

Cos site—a restriction site required of a small strand of DNA to be cleaved and packaged into the λ phage head.

Coupling (Bateson, Saunders, Punnett, 1905)—cis configuration (q.v.) of double heterozygotes.

Covalent bond—a stable chemical bond holding molecules together by sharing one or more pairs of electrons (as opposed to a noncovalent hydrogen bond).

Crossing-over (Morgan and Cattell, 1912)—the exchange of genetic information between two homologous chromosomes by chiasma formation (q.v.) in the diplotene stage of meiosis I; leads to genetic recombination of neighboring (linked) gene loci.

Crossing-over may also occur in somatic cells during mitosis (Stern, 1936). Unequal crossing-over (Sturtevant, 1925) results from mispairing of the homologous DNA segments at the recombination site. It results in structurally altered DNA segments or chromosomes, with a duplication in one and a deletion in the other.

Cyclic AMP (cAMP)—cyclic adenine monophosphate, a second messenger produced in response to stimulation of G-protein-coupled receptors.

Cyclin—a protein involved in cell cycle regulation.

Cytokine—a small secreted molecule that can bind to cell surface receptors on certain cells to trigger their proliferation or differentiation.

Cytoplasmic inheritance—transmission of genetic information located in mitochondria. Since sperm cells do not contain mitochondria, the transmitted information is of maternal origin.

Cytoskeleton—network of stabilizing protein in the cytoplasm and cell membrane.

Dalton—a unit of atomic mass, equal to 1/12 the mass of the of the ^{12}C nuclide ($1.657 \times 10^{-24}\text{gm}$).

Deficiency (Bridges, 1917)—loss of a chromosomal segment resulting from faulty crossing-over, e.g., by unequal crossing-over or by crossing-over within an inversion (q.v.) or within a ring chromosome (q.v.). It arises at the same time as a complementary duplication (q.v.). This is referred to as duplication/deficiency.

Deletion (Painter & Muller, 1929)—loss of part of or a whole chromosome or loss of DNA nucleotide bases.

Denaturation of DNA—separation of double-stranded nucleic acid molecules into single strands. Rejoining of the complementary single strands is referred to as renaturation.

Diakinesis (Haecker, 1897)—a stage during late prophase I of meiosis.

Dicentric (Darlington, 1937)—refers to a structurally altered chromosome with two centromeres.

Dictyotene—a stage of fetal oocyte development during which meiotic prophase is interrupted. In human females, oocytes attain the stage of dictyotene about 4 weeks before birth; further development of the oocytes is arrested until ovulation, at which time meiosis is continued.

Diploid (Strasburger, 1905)—cells or organisms that have two homologous sets of chromosomes, one from the father (paternal) and one from the mother (maternal).

Diplotene—a stage of prophase I of meiosis.

Discordance—the occurrence of a given trait or disease in only one member of a pair of twins.

Disomy, uniparental (UPD)—presence of two chromosomes of a pair from only one of the parents. One distinguishes UPD due to isodisomy, in which the chromosomes are identical, and heterodisomy, in which they are homologous.

Dispermy—the penetration of a single ovum by two spermatozoa.

Dizygotic—twins derived from two different zygotes (fraternal twins), as opposed to monozygotic (identical) twins, derived from the same zygote.

D loop—a region in mitochondrial DNA where one strand is paired with RNA, which displaces the other strand.

DNA (deoxyribonucleic acid)—the molecule containing the primary genetic information in the form of a linear sequence of nucleotides in groups of threes (triplets) (see codon).

Satellite DNA (sDNA) (Sueoka, 1961; Britten & Kohne, 1968)—contains tandem repeats of nucleotide sequences of different lengths. sDNA can be separated from the main DNA by density gradient centrifugation in cesium chloride, after which it appears as one or several bands (satellites) separated from that of the main body of DNA. In eukaryotes, light (AT-rich) and heavy (GC-rich) satellite DNA can be distinguished.

Microsatellites are small (2–10) tandem repeats of DNA nucleotides; minisatellites are tandem repeats of about 20–100 base pairs; classical satellite DNA consists of large repeats of 100–6500 bp (see p. 353).

DNA library—a collection of cloned DNA molecules comprising the entire genome (genomic library) or of cDNA fragments obtained from mRNA produced by a particular cell type (cDNA library).

DNA polymerase—a DNA-synthesizing enzyme. To begin synthesis, it requires a primer of RNA or a complementary strand of DNA.

DNase (deoxyribonuclease)—an enzyme that digests DNA.

Domain—a distinctive region of the tertiary structure of a protein or a particular region of a chromosome.

Dominant (Mendel, 1865)—refers to a genetic trait that can be observed in the heterozygous state. The terms “dominant” and “recessive” refer to the effects of the alleles at a given gene locus. The effects observed depend in part on the accuracy of observation. When the effects of each of two different alleles at a (heterozygous) locus can be observed, the alleles are said to be codominant. At the DNA level, allelic genes at two homologous loci are codominant.

Downstream—the 3' direction of a gene.

Drift, genetic (Wright, 1921)—random changes in gene frequency of a population. Especially relevant in small populations, where random differences in the reproductive frequency of a certain allele can change the frequency of the allele. Under some conditions an allele may disappear completely from a population (loss) or be present in all individuals of a population (fixation).

Duplication (Bridges, 1919)—addition of a chromosomal segment resulting from faulty crossing-over (see deficiency). It may also refer to additional DNA nucleotide base pairs. Duplication of genes (gene duplication) played an important role in the evolution of eukaryotes.

Electrophoresis (Tiselius, 1937)—separation of molecules by utilizing their different speeds of migration in an electrical field. As support medium, substances in gel form such as starch, agarose, acrylamide, etc. are used. Further molecular differences can be detected by modifications such as two-dimensional electrophoresis (electric field rotated 90° for the second migration) or cessation of migration at the isoelectric point (isoelectric focusing).

Elongation—addition of amino acids to a polypeptide chain.

Elongation factor—one of the proteins that associate with ribosomes while amino acids are added; EF in prokaryotes and eEF in eukaryotes.

Endocytosis—specific uptake of extracellular material at the cell surface. The material is surrounded by an invagination of the cell membrane, which pinches off to form a membrane-bound vesicle containing the material.

Endonuclease—a heterogeneous group of enzymes that cleave bonds between nucleotides of single- or double-stranded DNA or of RNA.

Endoplasmic reticulum—a complex system of membranes in the cytoplasm.

Endoreduplication (Levan & Hauschka, 1953)—chromosome replication during interphase without actual mitosis. Endoreduplicated chromosomes in metaphase consist of four chromatids lying next to each other, held together by two neighboring centromeres.

Enhancer—a cis-acting regulatory DNA segment that contains binding sites for transcription factors. An enhancer is located at various distances from the promoter. It causes an increase in the rate of transcription (about tenfold).

Enzyme (E. Büchner, 1897)—a protein that catalyzes a biochemical reaction. Enzymes consist of a protein part (apoenzyme), responsible for the specificity, and a nonprotein part (coenzyme), needed for activity. Enzymes bind to their substrates, which become metabolically altered or combined with other substances during the train of the reaction. Most of the enzymatically catalyzed chemical reactions can be classified into one of six groups:

- (1) hydrolysis (cleavage with the addition of H₂O), by *hydrolases*;
- (2) transfer of a molecular group from a donor to a receptor molecule, by *transferases*;
- (3) oxidation and reduction, by *oxidases* and *reductases* (transfer of one or more electrons or hydrogen atoms from a molecule to be oxidized to another molecule that is to be reduced);
- (4) isomerization, by *isomerases* (rearranging the position of an atom or functional group within a molecule);
- (5) joining of two substrate molecules to form a new molecule, by *ligases* (*synthetases*);

(6) nonhydrolytic cleavage with formation of a double bond on one or both of the two molecules formed, by *lyases*.

Epigenetic influence—a factor that changes the phenotype without altering the genotype.

Episome (Jacob & Wollman, 1958)—a plasmid (q.v.) that can exist either independently in the cytoplasm or as an integrated part of the genome of its bacterial host.

Epistasis (Bateson, 1907)—interaction of genes at the same gene locus (allelic) or at different gene loci (nonallelic) to alter phenotypic expression.

Epitope—the part of an antigen molecule that binds to an antibody.

EST (expressed sequence tag)—a sequenced site from an expressed gene that “tags” a stretch of unsequenced cDNA next to it; used to map genes (see STS).

Euchromatin (Heitz, 1928)—chromosome or chromosomal segment that stains less intensely than heterochromatin (q.v.). Euchromatin corresponds to the genetically active part of chromatin that is not fully condensed in the interphase nucleus.

Eukaryote (Chatton, 1925)—plants and animals with cells that have a chromosome-containing nucleus, which divides during mitosis and meiosis, in contrast to prokaryotes.

Euploid (Täckholm, 1922)—refers to cells, tissues, or individuals with the complete normal chromosomal complement characteristic of that species (cf. aneuploid, heteroploid, polyploid).

Excision repair—repair of bulk lesions in DNA in which a stretch of nucleotides (about 14 in prokaryotes and about 30 in eukaryotes) is excised from the affected strand and replaced by the normal sequence (resynthesis).

Exocytosis—specific process by which nondiffusible particles are transported through the cell membrane to be discharged into the cellular environment.

Exon (Gilbert, 1978)—a segment of DNA that is represented in the mature mRNA of eukaryotes (cf. intron).

Exonuclease—an enzyme that cleaves nucleotide chains at their terminal bonds only, either the 5' or 3' end (cf. endonuclease).

Expression—the observable effects of an active gene.

Expression vector—a cloning vector containing DNA sequences that can be transcribed and translated.

Expressivity (Vogt, 1926)—refers to the degree of phenotypic expression of a gene or genotype. Absence of expressivity is also called nonpenetrance.

Fibroblast—type of connective tissue cell. Can be propagated in culture flasks containing suitable medium (*fibroblast cultures*).

Fingerprint—a characteristic pattern of fragments of DNA or proteins.

Fitness, biological—refers to the probability (between 0.0 and 1.0) that a gene will be passed on to the next generation. For a given genotype and a given environment, the biological (or reproductive) fitness is determined by survival rate and fertility.

Founder effect—presence of a particular allele in a population due to a mutation in a single ancestor.

Gamete (Strasburger, 1877)—a haploid germ cell, either a spermatozoon (male) or an ovum (female). In mammals, males are heterogametic (XY) and females homogametic (XX). In birds, females are heterogametic (ZW) and males homogametic (ZZ).

Gene (Johannsen, 1909)—a hereditary factor that constitutes a single unit of hereditary material. It corresponds to a segment of DNA that codes for the synthesis of a single polypeptide chain (cf. cistron).

Gene amplification (Brown & David, 1968)—selective production of multiple copies of a given gene without proportional increases of other genes.

Gene bank—a collection of cloned DNA fragments that together represent the genome they are derived from (gene library).

Gene cluster (Demerec & Hartman, 1959)—a group of two or more neighboring genes of similar function, e.g., the HLA system or the immunoglobulin genes.

Gene conversion (Winkler, 1930; Lindgren, 1953)—interaction of alleles that leads to unequal genetic exchange during meiosis. Gene conversion refers to a process of nonreciprocal transfer of genetic information. One gene serves as a sequence donor, remaining unaffected, while the other gene receives sequences and undergoes variation.

Gene dosage—refers to the quantitative degree of expression of a gene. Also used to refer to the number of copies of a gene in the genome.

Gene family—a set of evolutionarily related genes by virtue of identity or great similarity of some of their coding sequences.

Gene flow (Berdseil, 1950)—transfer of an allele from one population to another.

Gene frequency—the frequency of a given allele at a given locus in a population (allele frequency).

Gene locus (Morgan, Sturtevant, Muller, Bridges, 1915)—the position of a gene on a chromosome.

Gene map—the position of gene loci on chromosomes. A physical map refers to the absolute position of gene loci, their distance from each other being expressed by the number of base pairs between them. A genetic map expresses the distance of genetically linked loci by their frequency of recombination.

Gene product—the polypeptide or ribosomal RNA coded for by a gene (see protein).

Genetic code—the information contained in the triplets of DNA nucleotide bases used to incorporate a particular amino acid into a gene product.

Genetic marker—a polymorphic genetic property that can be used to distinguish the parental origin of alleles.

Genetics (Bateson, 1906)—the science of heredity and the hereditary basis of organisms; derived from Gk. *genesis* (origin).

Genome (Winkler, 1920)—all of the genetic material of a cell or of an individual.

Genome scan—a search with marker loci on all chromosomes for linkage with an unmapped locus.

Genomics—the scientific field dealing with the structure and function of the entire genome (see part II, Genomics).

Genotype (Johannsen, 1909)—all or a particular part of the genetic constitution of an individual or a cell (cf. phenotype).

Germ cell—a cell able to differentiate into gametes by meiosis (as opposed to somatic cells).

Germinal—refers to germ cells, as opposed to somatic cells.

G6PD—glucose-6-phosphate dehydrogenase.

Gyrase—a topoisomerase that unwinds DNA.

Haploid (Strasburger, 1905)—refers to cells or individuals with a single chromosome complement; gametes are haploid.

Haplotype (Ceppellini et al., 1967)—a combination of alleles at two or more closely linked gene loci on the same chromosome, e.g., in the HLA system (q.v.).

Helicase—an enzyme that unwinds and separates the two strands of the DNA double helix by breaking the hydrogen bonds during transcription or repair.

Helix-loop-helix—a structural motif in DNA-binding proteins, such as some transcription factors.

Hemizygous—refers to genes and gene loci that are present in only one copy in an individual, e.g., on the single X chromosome in male cells (XY) or because the homologous locus has been lost.

Heritability (Lush, 1950; Falconer, 1960)—the ratio of additive genetic variance to the total phenotypic variance. Phenotypic variance is the result of the interaction of genetic and non-genetic factors in a population.

Heterochromatin (Heitz, 1928)—a chromosome or chromosomal segment that remains darkly stained in interphase, early prophase, and late telophase because it remains condensed, as all chromosomal material is in metaphase. This contrasts with euchromatin, which becomes invisible during interphase. Heterochromatin corresponds to chromosomes or chromosome segments showing little or no genetic activity. *Constitutive* and *facultative* heterochromatin

can be distinguished. An example of facultative heterochromatin is the heterochromatic X chromosome resulting from inactivation of one X chromosome in somatic cells of female mammals. An example of constitutive heterochromatin is the centric heterochromatin at centromeres that can be demonstrated as C bands.

Heterodisomy—presence of two homologous chromosomes from one parent only (cf. isodisomy and UPD).

Heteroduplex—refers to a region of a double-stranded DNA molecule with noncomplementary strands that originated from different duplex DNA molecules.

Heterogametic (Wilson, 1910)—producing two different types of gametes (q.v.), e.g., X and Y in (male) mammals or ZW in female birds.

Heterogeneity, genetic (Harris, 1953; Fraser, 1956)—an apparently uniform phenotype being caused by two or more different genotypes.

Heterokaryon (Ephrussi and Weiss, 1965; Harris and Watkins, 1965; Okada and Murayama, 1965)—a cell having two or more nuclei with different genomes.

Heteroploid (Winkler, 1916)—refers to cells or individuals with an abnormal number of chromosomes.

Heterosis (Shull, 1911)—increased reproductive fitness of heterozygous genotypes compared with the parental homozygous genotypes, in plants and animals.

Heterozygous (Bateson and Saunders, 1902)—having two different alleles at a given gene locus (cf. homozygous).

Hfr cell—a bacterium that possesses DNA sequences that lead to a high frequency of DNA transfer at conjugation.

HGPRT—hypoxanthine-guanine-phosphoribosyl transferase. An enzyme in purine metabolism that is inactive in Lesch-Nyhan syndrome.

Histocompatibility—tissue compatibility. Determined by the major histocompatibility complex MHC (see HLA).

Histone (Kossel, 1884)—chromosome-associated protein of the nucleosome. Histones H1-H4 form a nucleosome (q.v.).

HLA (Dausset, Terasaki, 1954)—human leukocyte antigen system A. HLA is also said by some to refer to Los Angeles, where Terasaki made essential discoveries.

Hogness box—a nucleotide sequence that is part of the promoter in eukaryotic genes.

Homeobox—a highly conserved DNA segment in homeotic genes.

Homeotic gene—one of the developmental genes in *Drosophila* that can lead to the replacement of one body part by another by mutation.

Homologous—refers to a chromosome or gene locus of similar maternal or paternal origin.

Homozygosity mapping—mapping genes by identifying chromosomal regions that are homozygous as a result of identity by descent from a common ancestor in consanguineous matings (see identity by descent, IBD).

Homozygous (Bateson and Saunders, 1902)—having identical alleles at a given gene locus.

Hox genes—clusters of mammalian genes containing homeobox sequences. They are important in embryonic development.

Hybridization—cross between two genotypically different plants or animals belonging to the same species. The term is often used in more narrow definitions: fusion of two single complementary DNA strands (DNA/DNA hybridization), fusion of complementary DNA and RNA strands (DNA/RNA hybridization), or the *in vitro* fusion of cultured cells of different species (cell hybridization).

Hydrogen bond—a noncovalent weak chemical bond between an electronegative atom (usually oxygen or nitrogen) and a hydrogen atom; important in stabilizing the three-dimensional structure of proteins or base pairing in nucleic acids.

Identity by descent (IBD)—refers to homozygous alleles at one gene locus that are identical because they are inherited from a common ancestor (see consanguinity).

Immunoglobulin — an antigen-binding molecule.

Imprinting, genomic—different expression of an allele or chromosomal segment depending on the parental origin.

Inbreeding coefficient (Wright, 1929)—measure of the probability that two alleles at a gene locus of an individual are identical by descent, i.e., that they are copies of a single allele of an ancestor common to both parents (IBD, identity-by-descent). Also, the proportion of loci at which the individual is homozygous.

Incidence—the rate of occurrence of a disease in a population. In contrast, prevalence is the percentage of a population that is affected with a particular disease at a particular time.

Inducer—a molecule that induces the expression of a gene.

Initiation factor—a protein that associates with the small subunit of a ribosome when protein synthesis begins (IF in prokaryotes, eIF in eukaryotes).

Insertion—insertion of chromosomal material of nonhomologous origin into a chromosome without reciprocal translocation (q.v.).

Insertion sequence (IS)—a small bacterial transposon carrying genes for its own transposition (q.v.).

In silico—a process taking place within a computer; for example, analysis of biological data.

Intercalating agent—a chemical compound that can occupy a space between two adjacent base pairs in DNA.

Interphase—the period of the cell cycle between two cell divisions (see mitosis).

Intron (Gilbert, 1978)—a segment of noncoding DNA within a gene (cf. exon). It is transcribed, but removed from the primary RNA transcript before translation.

Inversion (Sturtevant, 1926)—structural alteration of a chromosome through a break at two sites with reversal of direction of the intermediate segment and reattachment. A *pericentric inversion* includes the centromere in the inverted segment. A *paracentric inversion* does not involve the centromere. An inversion *per se* does not cause clinical signs, but it represents a potential genetic risk because crossing-over may occur in the region of the inversion and lead to aneusomy in offspring (aneusomy by recombination). Chromosomal inversions played an important role in evolution.

Inverted repeat—two identical, oppositely oriented copies of the same DNA sequence. They are an important feature of retroviruses.

In vitro—a biological process taking place outside a living organism or in an artificial environment in the laboratory.

In vivo—within a living organism.

Isochromatid break—see chromatid.

Isochromosome (Darlington, 1940)—a chromosome composed of two identical arms connected by the centromere, e.g., two long or two short arms of an X chromosome. Implies duplication of the doubled arm and deficiency of the absent arm. An isochromosome may have one or two centromeres.

Isodisomy—presence of two identical chromosomes from one of the parents (cf. heterodisomy).

Isolate, genetic (Waklund, 1928)—a physically or socially isolated population that has not interbred with individuals outside of that population (no panmixis).

Isozyme—closely related chains of immunoglobulins.

Isozyme or isoenzyme (Markert and Möller, 1959; Vesell 1959)—one of multiple distinguishable forms of enzymes of similar function in the same organism. Isoenzymes are a biochemical expression of genetic polymorphism.

Karyotype (Levitsky, 1924)—the chromosome complement of a cell, an individual, or a species.

Knockout (of a gene)—intentional inactivation of a gene in an experimental organism in order to obtain information about its function (same as targeted gene disruption).

Lagging strand of DNA—the new strand of DNA replicated from the 3' to 5' strand. It is synthesized in short fragments in the 5' to 3' direction (Okazaki fragments), which are subsequently joined together.

Lampbrush chromosome (Rückert, 1892)—a special type of chromosome found in the primary oocytes of many vertebrates and invertebrates during the diplotene stage of meiotic division and in drosophila spermatocytes. The chromosomes show numerous lateral loops of

DNA that are accompanied by RNA and protein synthesis.

Lariat—an intermediate form of RNA during splicing when a circular structure with a tail is formed by a 5' to 2' bond.

Leader sequence—a short N-terminal sequence of a protein that is required for directing the protein to its target.

Leaky mutant—a mutation causing only partial loss of function.

Leptotene—a stage of meiosis (q.v.).

Lethal equivalent (Morton, Crow, and Muller, 1956)—a gene or combination of genes that in the homozygous state is lethal to 100% of individuals. This may refer to a gene that is lethal in the homozygous state, to two different genes that each have 50% lethality, to three different genes each with 33% lethality, etc. It is assumed that each individual carries about 5–6 lethal equivalents.

Lethal factor (Bauer, 1908; Hadorn, 1959)—an abnormality of the genome that leads to death in utero, e.g., numerous chromosomal anomalies.

Library—see DNA library.

Ligand—a molecule that can bind to a receptor and thereby induce a signal in the cell, e.g., a hormone.

LINE (long interspersed nuclear element)—long interspersed repetitive DNA sequences (see p. 352).

Linkage disequilibrium (Kimura, 1956)—non-random association of alleles at closely linked gene loci that deviates from their individual frequencies as predicted by the Hardy-Weinberg equilibrium. Linkage disequilibrium is usually due to the founder effect (q.v.).

Linkage, genetic (Morgan, 1910)—localization of gene loci on the same chromosome close enough to cause deviation from independent segregation.

Linkage group—gene loci on the same chromosome which are so close together that they usually are inherited together without recombination.

Linker DNA—a synthetic DNA double strand that carries the recognition site for a restriction enzyme and that can bind two DNA fragments. Also, the stretch of DNA between two nucleosomes.

Locus—see gene locus.

Lymphocyte—cell of the immune system, of one of two general types: B lymphocytes from the bone marrow and thymus-derived T lymphocytes.

Lysosome (deDuve et al., 1955)—small cytoplasmic organelle containing hydrolytic enzymes.

Map distance—distance between gene loci, expressed in either physical terms (number of base pairs, e.g. kb 1000 bp or Mb million bp) or genetic terms (recombination frequency, expressed as cM, centimorgan. One cM corresponds to 1%).

Mapping—various methods to determine the position of a gene on a chromosome (physical map) or its relative distance to other gene loci and their order (genetic map).

Marker, genetic—an allele used to recognize a particular genotype.

Megabase (Mb)—1 million base pairs.

Meiosis (Farmer and Moore, 1905)—the special division of a germ cell nucleus that leads to reduction of the chromosome complement from the diploid to the haploid state. Prophase of the first meiotic division is especially important and consists of the following stages: leptotene, zygotene, pachytene, diplotene, diakinesis.

Mendelian inheritance (Castle, 1906)—inheritance according to the laws of Mendel as opposed to extrachromosomal inheritance, under the control of cytoplasmic hereditary factors (mitochondrial DNA).

Metabolic cooperation (Subak-Sharpe et al., 1969)—correction of a phenotype in cells in culture by contact with normal cells or cell products. An example of metabolic correction is the cross correction of cultured cells of different mucopolysaccharide storage diseases or correction of HGPRT-deficient cells by normal cells.

Metacentric—refers to chromosomes that are divided by the centromere into two arms of approximately the same length.

Metaphase (Strasburger, 1884)—stage of mitosis in which the contracted chromosomes are readily visible.

MHC (major histocompatibility complex) (Thorsby, 1974)—the principal histocompatibility system, consisting of class I and class III antigen genes of the HLA system including the class II genes.

Missense mutation—a mutation that alters a codon to one for a different amino acid (see nonsense mutation).

Mitosis (Flemming, 1882)—nuclear division during the division of somatic cells, consisting of prophase, metaphase, anaphase, and telophase.

Mitosis index (Minot, 1908)—the proportion of cells present that are undergoing mitosis.

Mixoploidy (Nemec, 1910; Hamerton, 1971)—a tissue or individual having cells with different numbers of chromosomes (chromosomal mosaic).

MLC—mixed lymphocyte culture (Bach and Hirschhorn, Bach and Lowenstein, 1964). MLC is a test for differences in HLA-D phenotypes.

Modal number (White, 1945)—the number of chromosomes of an individual or a cell.

Monoclonal antibody—an antibody representing a single antigen specificity, produced from a single progenitor cell.

Monolayer (Abercrombie and Heaysman, 1957)—the single-layered sheet of cultured diploid cells on the bottom of a culture flask.

Monosomy (Blakeslee, 1921)—absence of a chromosome in an otherwise diploid chromosomal complement.

Monozygotic—pertaining to uniovular twins (cf. dizygotic).

Morphogen—a protein present in embryonic tissues in a concentration gradient that induces a developmental process.

Mosaic—tissue or individuals made up of genetically different cells, as a rule of the same zygotic origin (cf. chimera).

mRNA (Brenner, Jacob, and Meselson, 1961; Jacob and Monod, 1961)—messenger RNA.

mtDNA—mitochondrial DNA.

Multigene family—a group of genes related by their common evolution.

Mutagen—a chemical or physical agent that can induce a mutation.

Mutation (de Vries, 1901)—permanent alteration of the genetic material. Different types include point mutations from exchange, loss, or insertion of base pairs within a gene and chromosomal mutation with alteration of the chromosome structure. A *missense mutation* is an alteration resulting in a gene product containing a substitution for a wrong amino acid. A *nonsense mutation* is an alteration that produces a stop codon in the midst of a genetic message so that a totally inadequate gene product is formed.

Mutation rate—the frequency of a mutation per locus per individual per generation.

Noncovalent bond—a (weak) chemical bond between an electronegative atom (usually oxygen or nitrogen) and a hydrogen atom (see hydrogen bond), but not involving sharing of electrons.

Nondisjunction (Bridges, 1912)—faulty distribution of homologous chromosomes at meiosis. In mitotic nondisjunction, the distribution error occurs during mitosis.

Nonsense codon—a codon that does not have a normal tRNA molecule. Any of the three triplets that terminate translation (UAG, UAA, UGA).

Nonsense mutation—a mutation that results in lack of any genetic information, e.g., a stop codon (see missense mutation).

Northern blot—transfer of RNA molecules to a membrane by a procedure similar to that of a Southern blot (q.v.).

Nucleoside—compound of a purine or a pyrimidine base with a sugar (ribose or deoxyribose) (cf. nucleotide).

Nucleosome (Navashin, 1912; Kornberg, 1974)—a subunit of chromatin consisting of DNA wound around histone proteins in a defined spatial configuration.

Nucleotide—Single monomeric building block of a polynucleotide chain that makes up nucleic acid. A nucleotide is a phosphate ester con-

sisting of a purine or a pyrimidine base, a sugar (ribose or deoxyribose as a pentose), and a phosphate group (see p. 28).

Ochre codon—the stop codon UAA.

Okazaki fragment—a short nucleotide sequence that is synthesized on the lagging strand of DNA during replication (q.v.) (see p. 42).

Oncogene (Heubner and Todaro, 1969)—a DNA sequence of viral origin that can lead to malignant transformation of a eukaryotic cell after being integrated into the cell genome (see proto-oncogene).

Open reading frame (ORF)—a DNA sequence of variable length that does not contain stop codons and therefore can be translated (see p. 48).

Operator (Jacob and Monod, 1959)—the recognition site of an operon at which the negative control of genetic transcription takes place by binding to a repressor.

Operon (Jacob et al., 1960)—in prokaryotes, a group of functionally and structurally related genes that are regulated together.

Origin of replication (ORI)—site of start of DNA replication.

Ortholog—a homologous DNA sequence or a gene that has evolved from a common ancestor between species, e.g., the α - and β -globin genes (see paralog).

Pachytene (de Winiwarter, 1900)—stage of prophase meiosis I.

Palindrome (Wilson and Thomas, 1974)—adjacent inverted repetitive linear base sequences of DNA that can help to form a hairpinlike structure by base-pairing of complementary sequences.

Panmixis (Weismann, 1895)—pairing system with random partner selection, as opposed to assortative mating.

Paralog—a DNA sequence or a gene that has evolved from a common ancestor within a species, e.g., the two α loci in humans (see ortholog).

Parasexual (Pontocorvo, 1954)—refers to genetic recombination by nonsexual means, e.g., by hybridization of cultured cells (see hybridization).

PCR (polymerase chain reaction) (Mullis 1985)—technique for in vitro propagation (amplification) of a given DNA sequence. It is a repetitive thermal cyclic process consisting of denaturation of genomic DNA of the sequence of interest, annealing the DNA to appropriate oligonucleotide primers, and replication of the DNA segment complementary to the primer.

Penetrance (Vogt, 1926)—the frequency or probability of expression of an allele (cf. expressivity).

Peptide—a compound of two or more amino acids joined by peptide bonds.

Phage—see bacteriophage.

Phenocopy (Goldschmidt, 1935)—a nonhereditary phenotype that resembles a genetically determined phenotype.

Phenotype (Johannsen, 1909)—the observable effect of one or more genes on an individual or a cell.

Pheromone—a signaling molecule that can alter the behavior or gene expression of other individuals of the same species.

Phosphodiester bond—the chemical bond linking adjacent nucleotides of DNA or RNA.

Phytohemagglutinin (PHA)—a protein substance obtained from kidney beans (*Phaseolus vulgaris*). It is used to separate red from white blood cells. Nowell (1960) discovered its ability to induce blastic transformation (see Transformation) and cell division in lymphocytes. It is the basis of phytohemagglutinin-stimulated lymphocyte cultures for chromosomal analysis.

Plasmid (Lederberg, 1952)—autonomously replicating circular DNA structures found in bacteria. Although they are usually separate from the actual genome, they may become integrated into the host chromosome.

Plastids—any of several types of organelles found in plant cells, e.g., chloroplasts.

Pleiotropy (Plate, 1910)—expression of a gene with multiple, seemingly unrelated phenotypic features.

Point mutation—alteration of the genetic code within a single codon. The possible types are the exchange of a base: a pyrimidine for another pyrimidine (or a purine for another purine) as a

transition (Frese, 1959), i.e., thymine for cytosine (or adenine for guanine); or the exchange of a pyrimidine by a purine or visa versa: *transversion*, i.e., thymine by adenine or visa versa (Frese, 1959). Besides the two types of exchange, a point mutation may be due to the insertion of a nucleotide base or the deletion of one or several base pairs.

Polar body (Robin, 1862)—an involutinal cell arising during oogenesis that does not develop further as an oocyte.

Polyadenylation—the addition of multiple adenine residues at the 3' end of eukaryotic mRNA after transcription.

Polycistronic messenger—mRNA including coding regions from more than one gene (in prokaryotes).

Polygenic (Plate, 1913; Mather, 1941)—refers to traits that are based on several or numerous genes whose effects cannot be individually determined. The term multigenic is sometimes used instead.

Polymerases—enzymes that catalyze the combining of nucleotides to form RNA or DNA (genetic transcription and DNA replication).

Polymorphism, genetic (Ford, 1940)—existence of more than one normal allele at a gene locus, with the rarest allele exceeding a frequency of 1%. A polymorphism may exist at several levels, i.e., variants in DNA sequence, amino acid sequence, chromosomal structure, or phenotypic traits (pp. 164 ff).

Polypeptide—see peptide.

Polyploid (Strasburger, 1910)—refers to cells, tissues, or individuals having more than two copies of the haploid genome, e.g., three (triploid) or four (tetraploid). In man, triploidy and tetraploidy in a conceptus are usually lethal and as a rule lead to spontaneous abortion.

Polytene (Koltzoff, 1934; Bauer, 1935)—refers to a special type of chromosome resulting from repeated endoreduplication of a single chromosome. Giant chromosomes arise in this manner (cf. chromosome).

Population (Johannsen, 1903)—individuals of a species that interbreed and constitute a common gene pool (cf. race).

Positive negative effect—a mutation exerting a negative functional effect in one allele.

Premature chromosomal condensation (Johnson and Rao, 1970)—induction of chromosomal condensation in an interphase nucleus after fusion with a cell in mitosis. Condensed S-phase chromosomes appear pulverized (so-called chromosomal pulverization).

Prevalence—see incidence.

Pribnow box—part of a promoter (TATAAT sequence 10 bp upstream of the gene) in prokaryotes.

Primary transcript—the original RNA transcribed from a eukaryotic gene before processing (splicing, addition of the cap, and polyadenylation).

Primer—a DNA or RNA oligonucleotide that after hybridization to an inversely complementary DNA has a 3'-OH end to which nucleotides can be added for synthesis of a new chain by DNA polymerase.

Prion—proteinaceous infectious particles that cause degenerative disorders of the central nervous system.

Proband—see Propositus.

Probe—a defined DNA or RNA fragment used to identify complementary sequences by specific hybridization.

Prokaryote—one-celled organism without a cell nucleus or intracellular organelles.

Promoter—a defined DNA region at the 5' end of a gene that binds to transcription factors and RNA polymerase during the initiation of transcription. The -10 sequence is the consensus sequence TATAATG about 10 bp upstream of a prokaryotic gene (Pribnow box).

Prophage—a viral (phage) genome integrated into the bacterial (host) genome.

Propositus, proband—the individual in a pedigree that has brought a family to attention for genetic studies.

Protein—one or more polypeptides with a specific amino acid sequence and a specific three-dimensional structure. Proteins are biomolecules representing the key structural elements of living cells and participating in nearly

all cellular and biochemical reactions (see gene product).

Proteome—the complete set of all protein-encoding genes or all the proteins produced by them.

Proto-oncogene (cellular oncogene)—a eukaryotic gene. It may be present in truncated form in a retrovirus, where it may behave as an oncogene.

Prototrophic—refers to cells or cell lines that do not require a special nutrient added to the culture medium (cf. auxotrophic).

Provirus—duplex DNA derived from an RNA retrovirus and incorporated into a eukaryotic genome.

Pseudogene—DNA sequences that closely resemble a gene but are without function due to an integral stop codon, deletion, or other structural change. A processed pseudogene consists of DNA sequences that resemble the mRNA copy of the parent gene, i.e., it does not contain introns.

Pseudohermaphroditism—a condition in which an individual has the gonads of one sex and phenotypic features of the opposite sex.

Quadriradial figure—the configuration assumed when homologous segments of chromosomes involved in a reciprocal translocation pair at meiosis. Rarely, such a figure may occur during mitosis.

Race—a population (q.v.) that differs from another population in the frequency of some of its gene alleles (L. C. Dunn: *Heredity and Evolution in Human Populations*, Harvard University Press, Cambridge, Mass., 1967). Accordingly, the concept of race is flexible and relative, defined in relation to the evolutionary process. The term race can be used to classify groups, whereas the classification of individuals is often uncertain and of biologically dubious value.

Reading frame—sequence of DNA nucleotides that can be read in triplets to code for a peptide (cf. open reading frame).

Receptor—a transmembrane or intracellular protein involved in transmission of a cell signal.

Recessive (Mendel, 1865)—refers to the genetic effect of an allele (q.v.) at a gene locus that is

manifest as phenotype in the homozygous state only (q.v.).

Reciprocal translocation—mutual exchange of chromosome parts.

Recombinant DNA—A DNA molecule consisting of parts of different origin.

Recombination (Bridges and Morgan, 1923)—the formation of new combinations of genes as a result of crossing-over between homologous chromosomes during meiosis.

Recombination frequency—frequency of recombination between two or more gene loci. Expressed as the theta (θ) value. A θ of 0.01 (1% recombination frequency) corresponds to 1 centimorgan (cM).

Regulatory gene—a gene coding for a protein that regulates other genes.

Renaturation of DNA—combining of complementary single strands of DNA to form double-stranded DNA (cf. denaturation).

Repair (Muller, 1954)—correction of structural and functional DNA damage.

Replication—identical duplication of DNA.

Replication fork—the unwound region of the DNA double helix in which replication takes place.

Replicon (Huberman and Riggs, 1968)—an individual unit of discontinuous DNA replication in eukaryotic DNA.

Reporter gene—a gene used to analyze another gene, especially the regulatory region of the latter.

Repulsion (Bateson, Saunders, and Punnett, 1905)—term to indicate that the mutant alleles of neighboring heterozygous gene loci lie on opposite chromosomes, i.e., in trans configuration (see cis/trans).

Resistance factor—a plasmid gene causing antibiotic resistance.

Restriction enzyme, or restriction endonuclease (Meselson and Yuan, 1968)—endonuclease that cleaves DNA at a specific base sequence (restriction site or recognition sequence).

Restriction map—a segment of DNA characterized by a particular pattern of restriction sites.

Restriction site—a particular sequence of nucleotide bases in DNA that allow a particular restriction enzyme to cleave the DNA molecule at, or close to, that site (recognition site).

Retrotransposon—a mobile DNA sequence that can insert itself at a different position by using reverse transcriptase (see transposon).

Retrovirus—a virus with a genome consisting of RNA that multiplies in a eukaryotic cell by conversion into duplex DNA.

Reverse transcriptase—an enzyme complex that occurs in RNA viruses and that can synthesize DNA from an RNA template.

RFLP—restriction fragment length polymorphism. The production of DNA fragments of different lengths by a given restriction enzyme, due to inherited differences in a restriction site.

Rho factor—a protein involved in termination of transcription in *E. coli*.

Ribosome (Roberts, 1958; Dintzis et al., 1958)—complex molecular structure in pro- and eukaryotic cells consisting of specific proteins and ribosomal RNA in different subunits. The translation of genetic information occurs in ribosomes.

Ring chromosome—a circular chromosome. In prokaryotes the normal chromosome is ring-shaped. In mammals it represents a structural anomaly and implies that chromosomal material has been lost.

RNA (ribonucleic acid)—a polynucleotide with a structure similar to that of DNA except that the sugar is ribose instead of deoxyribose.

Satellite (Navashin, 1912)—small mass of chromosomal material attached to the short arm of an acrocentric chromosome (q.v.) by a constricted appendage or stalk. It is involved in the organization of the nucleolus. The stalk region can be stained by specific silver stain (NOR stain, nucleolus-organizing region). The size of the satellite, the length of its stalk, and the intensity of its fluorescence after staining with acridine are polymorphic cytogenetic markers.

Satellite DNA (sDNA) (Sueoka, 1961; Kit, 1961; Britten and Kohne, 1968)—DNA that is either heavier (GC-rich) or lighter (AT-rich) than the main DNA (see DNA). Not to be confused with the satellite regions of acrocentric chromosomes.

SCE (sister chromatid exchange) (Taylor, 1958)—an exchange between the two chromatids of a metaphase chromosome. After two replication cycles in a cell culture in the presence of a halogenated base analog (e.g., 5-bromodeoxyuridine), both DNA strands of one chromatid will be substituted with the halogenated base analog, whereas only one DNA strand of the other chromatid will be substituted. As a result, the two chromatids differ in staining intensity, and it is possible to determine where crossing-over of the two chromatids has occurred (see p. 334).

Segregation (Bateson and Saunders, 1902)—the separation of alleles at a gene locus at meiosis and their distribution to different gametes. Segregation accounts for the 1 : 1 distribution of allelic genes to different chromosomes.

Selection (Darwin, 1858)—preferential reproduction or survival of different genotypes under different environmental conditions.

Selection coefficient—quantitative expression (from 0 to 1) of the disadvantage that a genotype has (compared with a standard genotype) in transmitting genes to the next generation. The selection coefficient (s) is the numerical amount by which biological fitness ($1-s$) is decreased; i.e., a selection coefficient of 1 indicates a complete lack of biological fitness.

Selective medium—a medium that supports growth of cells in culture containing a particular gene.

Semiconservative (Delbrück and Stent, 1957)—the normal type of DNA replication. One DNA strand is completely retained; the other is synthesized completely anew.

Signal sequence—the N-terminal amino acid sequence of a secreted protein; required for transport of the protein to the right destination in the cell.

Silencer sequence—a eukaryotic DNA sequence that blocks the access of gene activity proteins required for transcription, by forming condensed heterochromatin in that particular area.

SINE (short interspersed nuclear element)—short repetitive DNA sequences (cf. LINE).

snRNPs (small nuclear ribonucleoprotein particles)—complexes of small nuclear RNA molecules and proteins.

Somatic—refers to cells and tissues of the body, as opposed to germinal (referring to germ cells).

Somatic cell—any cell of an organism that does not undergo meiosis and does not form gametes (as opposed to germ cells).

Somatic cell hybridization—formation of cell hybrids in culture.

Southern blot (Southern, 1975)—method of transferring DNA fragments from an agarose gel to a membrane after the fragments have been separated according to size by electrophoresis.

Speciation (Simpson, 1944)—formation of species during evolution. One of the first steps toward speciation is the establishment of a reproductive barrier against genetic exchange. A frequent mechanism is chromosomal inversion.

Species (Ray, 1670)—a natural population in which there is interbreeding of the individuals, which share a common gene pool.

S phase (Howard and Pelc, 1953)—phase of DNA synthesis (DNA replication) between the G1 and the G2 phase of the eukaryotic cell cycle.

Spliceosome—an aggregation of different molecules that can splice RNA.

Splicing—a step in processing a primary RNA transcript in which introns are excised and exons are joined.

Splice junction—the sequences at the exon/intron boundaries.

Stem cell—a cell able to renew itself by division, retaining the potential for differentiation within a particular developmental pathway. One distinguishes omnipotent and pluripotent stem cells.

STS (sequence tagged site)—a short segment of DNA of known sequence.

Submetacentric—refers to a chromosome consisting of a short and a long arm because of the position of its centromere.

Synapsis (Moore, 1895)—the pairing of homologous chromosomes during meiotic prophase.

Synaptonemal complex (Moses, 1958)—parallel structures associated with chiasmata formation during meiosis, visible under the electron microscope.

Syndrome—within human genetics, a group of clinical and pathological characteristics that are etiologically related, regardless of whether the details of their relationship have yet to be identified.

Syteny (Renwick, 1971)—refers to gene loci that are located on the same chromosome, whether or not they are linked.

Tandem duplication—short identical DNA segments adjacent to each other.

TATA box—a conserved, noncoding DNA sequence about 25 bp in the 5' region of most eukaryotic genes. It consists mainly of sequences of the TATAAAA motif. Also known as Hogness box (cf. Pribnow box, of prokaryotes).

T cells—T lymphocytes.

Telocentric (Darlington, 1939)—refers to chromosomes or chromatids with a terminally located centromere, without a short arm or satellite. They do not occur in man.

Telomerase—a ribonucleoprotein enzyme that adds nucleotide bases at the telomere.

Telomere (Muller, 1940)—the terminal areas of both ends of a chromosome containing specific consensus sequences (see p. 180).

Template—the molecule that determines the nucleotide sequence for the formation of another, similar (complementary) molecule (see DNA and RNA).

Teratogen (Ballantyne, 1894)—chemical or physical agent that leads to disturbances of embryological development and malformations.

Termination codon—one of the three triplets signaling the end of translation (UAG, UAA, UGA).

Terminator—a DNA sequence that signals the end of transcription.

Tetraploid (Nemec, 1910)—having a double diploid chromosome complement, i.e., four of each kind of chromosome are present ($4n$ instead of $2n$).

Topoisomerase—a class of enzymes that can control the three-dimensional structure of DNA by cutting one DNA strand, rotating it about the other, and resealing it (class I) or cutting and re-

sealing both ends (class II). Used to unwind the DNA double helix at transcription.

Transcript—an RNA copy of a segment of the DNA of an active gene.

Transcription—the synthesis of messenger RNA (mRNA), the first step in relaying the information contained in DNA.

Transcription unit—all of the DNA sequences required to code for a given gene product (operationally corresponding to a gene). Includes promoter and coding and noncoding sequences.

Transduction (Zinder and Lederberg, 1952)—transfer of genes from one cell to another (usually bacteria) by special viruses, the bacteriophages.

Transfection—introduction of pure DNA into a living cell (cf. transformation).

Transformation—this term has several different meanings in biology. In genetics, three main types of transformation are distinguished: 1) *malignant transformation*, the transition of a normal cell to a malignant state with loss of control of proliferation; 2) *genetic transformation* (Griffith, 1928; Avery et al. 1944), a change of genetic attributes of a cell by transfer of genetic information; and 3) *blastic transformation*, the reaction of lymphocytes to mitogenic substances (e.g., phytohemagglutinin or specific antigens) leading to cell division.

Transgenic—refers to an animal or a plant into which a cloned gene has been introduced and stably incorporated. It reveals information about the biological function of the (trans)gene.

Transition—Replacement of a purine with another purine or a pyrimidine with another pyrimidine (see mutation).

Translation—the second step in the relay of genetic information. Here the sequence of triplets in mRNA are translated into a corresponding sequence of amino acids to form a polypeptide as the gene product.

Translocation—transfer of all or part of a chromosome to another chromosome. A translocation is usually *reciprocal*, leading to an exchange of nonhomologous chromosomal segments. A translocation between two acrocentric chromosomes that lose their short arms and fuse at their centromeres is called a *fusion* type translocation (Robertsonian translocation).

Transposon—a DNA sequence with the ability to move and become inserted at a new location of the genome.

Transversion—replacement of a purine with a pyrimidine or vice versa (see mutation).

Triplet—a sequence of three nucleotides comprising a codon of a nucleic acid and representing the code for an amino acid (triplet code, see codon).

Trisomy (Blakeslee, 1922)—an extra chromosome, in addition to a homologous pair of chromosomes.

UPD—uniparental disomy (see disomy).

Upstream—5' direction of a gene.

Variation—the differences among related individuals, e. g., parents and offspring, or among individuals in a population.

Variegation—different phenotypes within a tissue.

Vector—a molecule that can incorporate and transfer DNA.

Virion—a complete extracellular viral unit or particle.

Virus—DNA or RNA of defined size and sequence, enclosed in a protein coat encoded by its genes and able to replicate in a susceptible host cell only (see Part I).

Western blot—technique to identify protein antigens, in principle similar to the Southern blot method (q.v.).

Wild-type—refers to the genotype or phenotype of an organism found in nature or under standard laboratory conditions, roughly meaning "normal."

X chromatin (formerly called Barr body or sex chromatin) (Barr and Bartram, 1949)—darkly staining condensation in the interphase cell nucleus representing an inactivated X chromosome

Xenogenic—refers to transplantation between individuals of different species.

X-inactivation (Lyon, 1961)—inactivation of one of the two X chromosomes in somatic cells of female mammals during the early embryonic period by formation of X chromatin.

X-linked—refers to genes on the X chromosome.

YAC (yeast artificial chromosome)—a yeast chromosome into which foreign DNA has been inserted for replication in dividing yeast cells. YACs can incorporate relatively large DNA fragments, up to about 1000 kb.

Y chromatin (F body; Pearson, Bobrow, Vosa, 1970)—the brightly fluorescing long arm of the Y chromosome visible in the interphase nucleus.

Z DNA—alternate conformation of DNA. Unlike normal B DNA, (Watson-Crick model), the helix is left-handed and angled (zigzag, thus Z DNA).

Zinc finger—A finger-shaped region found in many DNA-binding regulatory proteins, the "finger" being held together by a strategically placed zinc atom.

Zoo blot—a Southern blot containing conserved DNA sequences from related genes of different species. It is taken as evidence that the sequences are coding sequences from a gene (see p. 250).

Zygote (Bateson, 1902)—the new diploid cell formed by the fusion of the two haploid gametes, an ovum and a spermatozoon, at fertilization. The cell from which the embryo develops.

Zygotene (de Winiwarter, 1900)—a stage of prophase of meiosis I.

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Glossary of Genetic Terms, National Institute of Human Genome Research
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